

 Gekakonus®		OPERATING INSTRUCTIONS NATURAL CIRCULATION STEAM BOILER TYPE NUK-HP	
SAFETY VALVE (S), RUPTURE DISC AND CONTACT MANOMETER			
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Appendix F

Safety valve(s), rupture disc and contact manometer

Contents:

Drawing no. 152466 A Mounting of safety valve

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1.0 SAFETY VALVES

1.1 FUNCTION

Every pressurised vessel has to be equipped with a safety valve to provide, that the system will run at a higher pressure than the max. allowable pressure p₀.

1.2 TECHNICAL DATA

Type test approved as full lift safety valve for steam/gases

Appr. No.: TÜV-SV 98-934 D/G/F

Coefficient of discharge for steam/gases $\alpha_d=0,8$

Body material 1.0619/ WCB

Bonnet material 1.0619/ WCB

Flange connection

Inlet:

DN 25 PN 160, DIN 2638

Outlet:

DN 50 PN 40, DIN 2635

1.3 INSTALLATION

General notes

The quality of the sealing surfaces is endangered by the ingress of foreign materials. Impurities must be prevented from entering the valve during transportation, installation and operation.

When installing safety valves with threaded connections, use metal seal washers only. Seal materials such as hemp or PTFE tape should not be used as his type of material can break off and enter the valve causing it to leak.

If valves with open bonnets and / or levers are to be painted, care must be taken to protect the sliding parts; otherwise the correct operation of the safety valves may be affected.

Transport Protection

All protective devices at transport and handling have to be removed before mounting the safety valve.

The lever of spring loaded safety valves is secured with a wire to the bonnet subject to the pressure setting. This wire should be removed only after installation. For testing the set pressure, or tightness of the safety valve, remove this wire and take care that the lever is not engaged with the spindle cap.

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Valves supplied with lever and weight are packed with a wooden wedge to secure the lever and prevent movement of the disc. This packing precaution should only be removed after installation.

Assembly

All drainage pipes that have been connected must be free of restriction and the ends easily observed. Care must be taken when discharging any drainage to prevent injury to personnel (i.e. on steam lines connect a suitable steam trap).

Attention! When valves are supplied with a drain hole that is not going to be used the plastic plug fitted in the hole must be removed and replaced by a metal plug.

The discharge piping on steam lines should be installed in an ascending direction, and on liquid lines in a descending direction. Spring loaded safety valves should be installed with the bonnet VERTICALLY UPWARDS.

Safety valves should be mounted in such a way that no inadmissible static, dynamic or thermal loads can be transmitted to the valve due to up or downstream pipe work. If necessary expansion devices should be utilised. Additional stresses by incorrect mounting should be avoided. Freezing and sticking of fluid or condensate in the valve body should be avoided by suitable actions.

To prevent the ingress of dirt and other impurities from entering the safety valve, drainage of the discharge pipe and safety valve must be implemented via the discharge pipe. Therefore, safety valves are not provided with drain holes.

According to technical regulations, a drain hole of sufficient size must be incorporated at the lowest point of pipe work. The discharge pipe must slope in a downward direction and be fitted with a drainpipe (which should be DN 25 minimum for discharge pipes DN 40 and larger) before any bends are connected.

Set point adjustment

The pressure setting is wire-locked and sealed against unauthorised alterations. TÜV regulations, agreed by the manufacturer, require the fitting of a type test approval plate stating the correct valve data. The manufacturer can not be held responsible for any changes to set pressure or other alterations after despatch from the factory. In case of any necessary modifications, we recommend to return the valve to the supplier, an authorised repair shop or to do the modifications under the supervision of the TÜV or any other suitable inspecting authority.

Lifting device H3 (See figure 1 below)

- Step 1. Remove shaft (45)
- Step 2. Pull lever (43) from lever cover
- Step 3. Loosen screw (84)
- Step 4. Unscrew and remove lever cover (41)
- Step 5. Slacken lock nut (19)
- Step 6. Turn adjusting screw (18) to the required set pressure

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Note: Draw attention to the admissible pressure range of the spring!

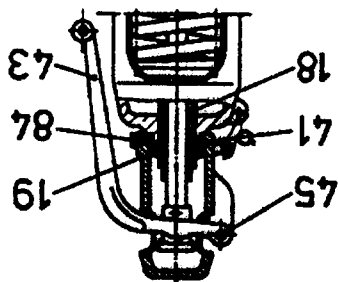


Figure 1 Lifting device H3

Turning the adjusting screw (18) in a clockwise direction increases the spring tension, thus obtaining a higher set point pressure. Turning the adjusting screw anticlockwise reduces the spring tension, thus obtaining a lower set point pressure.

Upon completion of adjustment, reassembly is in reverse order of steps 1 – 6 above, this locks out the set pressure.

1.4 INSPECTION AND TESTING

The safety valves are tested, adjusted to the required set pressure and sealed (see also DIN 3320). The set pressure and the LESER sign is marked on the edge of the outlet flange.

If the TÜV or any other licensed authority inspects the valves, their signs are also hard stamped on the edge of the flange or on the valve body.

The valve should be closed by sudden release of the lever. If this procedure does not stop the leakage, it is likely that the valve seat is damaged. Repairs should be carried out at GEKAKONUS GMBH 's workshop or by an authorised person.

It is recommended to lift the valve lever manually from time to time in order to prevent accumulation of deposits that will affect valve operation (testing of safety valves is required for plants operating with demineralised water and for hot water generators at intervals of at least six months).

It is important to ensure that the lifting lever does not engage the spindle cap after the lift operation. The lever must be deflected towards the centre-line of the bonnet until the lifting fork is disengaged.

1.5 OPERATION AND SAFETY

The working pressure of the plant should be at least 5% below the reseating pressure of the valve to enable a correct re-seat. The valve should be closed by sudden release of the lever. If

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this procedure does not stop the leakage, probably the valve seats are damaged. Repairs should be carried out at our workshop.

1.6 MAINTENANCE

It is recommended and in Germany required by rules, to lift manually the valves lifting lever from time to time in order to prevent accumulation of deposits that will affect the valve operation.

2.0 RUPTURE DISC

2.1 FUNCTION

A rupture disc has to be installed to provide leakages of the system, as the safety valve begins to open in a range of abt. 3-5 % below the set pressure.

2.2 TECHNICAL DATA

Type: RLS-90

Size: DN 25

Material: 1.4401

For more details see the "Burst Test Certificate".

2.3 INSTALLATION

The rupture disc will be installed boiler connection flange and the inlet connection of the safety valve, for details see drawing no. 152466 in this appendix.

2.4 INSPECTION AND TESTING

Check knife blades in outlet flange for sharpness. Dullness may affect rupture disc performance. If slightly dull or nicked, sharpen with mill file or stone. If nicks cannot be filed out, then replace flange. When penetration points are attached to the knife blades, then replace outlet flange. When penetration points are attached to the knife blades, they must be sharp and not bend.

2.5 OPERATION

During the normal operation the rupture disk is not damaged.

2.6 MAINTENANCE

Corrosion and service conditions may affect disk life, thus requiring periodic change. Regular disk inspection is advised every 3 month to insure that corrosion, creep, etc. are not a threat to predictable performance.

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3.0 CONTACT MANOMETER

3.1 FUNCTION

The contact manometer is used to monitor the pressure between the rupture disk and the safety valve.

3.2 TECHNICAL DATA

- Name: Contact manometer
- Type: 237 001
- Contact: 1 opener as magnet spring-contact
- Range: 0-100 bar
- Set point: 10 bar
- Casing diameter: 100 mm
- Indication accuracy: class 1,0
- Excess pressure safety: max. 130 bar short-term
- Contact load: 380 V - 30 W / 50 VA
- 380 V - 80 mA
- 220 V- 135 mA
- 110 V - 270 mA
- Contact material: Silfram W 30 = Silver + 30 % Wolfram
- Ambient temperature: -20.....+140°C
- Type of switch: W1/101
- Connection: R 1/4", below

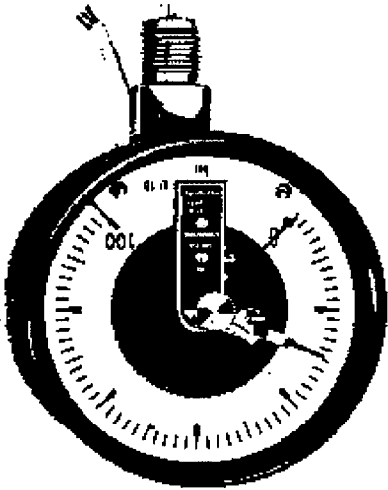


Figure 2 Contact manometer

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3.3 INSTALLATION

For installation details see drawing no. 152466 B in this appendix.

3.4 INSPECTION AND TESTING

The contact manometer could be tested via the testing cock at the front side see drawing no. 152466 B

3.5 OPERATION

Because of the rupture disc the manometer is not pressurised, the contact is closed. By auxiliary relays with indication lamps or auxiliary electric circuits, normal system operation could occur.

Activation of the rupture disc will mean that the full pressure is lying on the manometer, the contact will open and the related trouble indication lamp signal "excess pressure rupture disk". If this alarm is given the rupture disk is broken and has to be replaced.

3.6 MAINTENANCE

No special maintenance is required.

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Appendix G

Installation instructions for NUK HP systems

Drawing no. 154246 C
Example for pipe work NUK-HP

Drawing no. 154546 B
Max. admissible forces

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1.0 General

It is the responsibility of the owner (client) of NUK HP systems to ensure that an acceptable standard of workmanship is achieved at the time of the site installation.

Due diligence must be given to the quality of site fabrication, particularly the welding, because it cannot be stressed enough that the operating system must be pressure tight to ensure reliable service.

1.1 Design

A suitable nationally or internationally recognised code of construction should be adopted for the design and construction requirements of high pressure steam and condensate pipe work.

Such a code could be: ASME/ANSI B3.1 – Power piping.

The guidelines listed below must also be incorporated into the final system design layout.

Steam and condensate lines in working order must be installed with a constant slope of at minimum 5% from the heat consumers to the NUK. (i.e. if a contra-slope should develop due to the thermal expansion of the pipe, this can result in low water problems and strong steam hammering.

At operational temperatures (300 °C) the steam and condensate lines expand by 3.8 mm per meter (at 10 m long pipe sections by 38 mm).

With regard to the change of length, the steam and condensate lines must be designed such that the as-built construction includes sufficient length for compensation. Fix- and sliding points for the rising reaction power must be appropriately calculated.

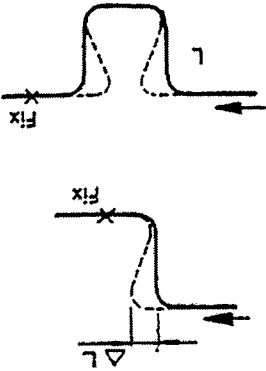
The laying of pipe-work must happen with adequate pre-stressing - i.e. half of the change of length must be installed in cold condition - opposite to the expansion direction.

Example of pipework compensation:

Expansion absorption: at L1 of 50 mm results a leg length L of:

- For a tube 42,4 x 2,9 = 5 m
- For a tube 60,3 x 3,2 = 6,5 m
- For a tube 76,1 x 3,6 = 8 m

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Pipework engineering has to be done by competent companies, the complete installation has to be accepted by the local authorities.

1.2 Materials of construction

Tube diameter:	> DN 65, St. 35.8, Quality I,	ASME, SA 192 or SA 220
	> DN 65, St. 35.8, Quality III	ASME, SA 192 or SA 220

Welding material autogenous: Material no. 1.6215

Welding material electro-arc welding: Gricon 8 A or similar

1.3 Inspection and testing

The non-destructive testing requirements stated in the relevant code of construction should be implemented on the pipework.
As a minimum this should include:

- 100% Visual examination on all weld connections
- 10% Radiographic examination on randomly selected (completed) butt welds
- 10% Examination of welding seams by dye penetration test

Upon completion, the system cleaning and pressure testing procedures stated in chapter 3 of this manual should be implemented and the appropriate certification issued (if satisfactory).

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The recommendation is based on following average operating conditions:

You must use the size of pipe dimension that you have calculated according to chapter 3 (page5/9). This required minimal pipe dimension is based on following average operating conditions:

Saturated steam
Operating pressure
Evaporating heat
Volume specific
Steam velocity
 $p = 50 \text{ bar}$
 $r = 390,0 \text{ kcal/kg}$
 $v'' = 0,03942 \text{ m}^3/\text{kg}$
 $< 10 \text{ m/s}$

Tubes acc. to DIN 17175 or 17155					Tubes acc. to ASA B 36.10. Schd. no. 80				
Boiler type	Tube (*)	Tube dimension	Tube content	Tube weight insul. and filled with water [kg/m]	Tube dimension	Tube content	Tube weight insul. and filled with water [kg/m]	Insul thickness [mm]	Heat loss [W/m]
NUK-HP	DN	d x s							
320	40	48,3x3,6	1,33	17	1 1/2"	48,3x5,08	1,14	18,5	87
435	40	48,3x3,6	1,33	17	1 1/2"	48,3x5,08	1,14	18,5	87
700	50	60,3x4,5	2,15	22	2,0"	60,3x5,54	1,9	23	92
930	50	60,3x4,5	2,15	22	2,0"	60,3x5,54	1,9	23	92
1250	65	76,1x5,0	3,43	27	2 1/2"	73,0x7,01	2,73	30	98
1500	65	76,1x5,0	3,43	27	2 1/2"	73,0x7,01	2,73	30	98
1745	80	88,9x6,3	4,57	32	3,0"	88,9x7,62	4,26	34	107
2100	100	114,3x8,0	7,59	42	4,0"	114,3x8,56	7,61	43,5	110
2325	100	114,3x8,0	7,59	42	4,0"	114,3x8,56	7,61	43,5	110
3500	100	114,3x8,0	7,59	42	4,0"	114,3x8,56	7,61	43,5	110
4200	125	139,7x10,0	11,71	66,0	5,0"	141,3x9,52	11,74	69,0	120
5350	150	168,3x12,5	16,12	96,0	6,0"	168,3x11,0	16,80	90,0	135

In case it should be not possible in practice, to install the calculated steam- and condensate line, please chose a larger tube diameter which shows a sufficient length for the real installation.
*) These tube dimensions are minimal tube dimension that must not be smaller

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Calculation sheet for the inside diameter of the steam- and condensate connection as well as length

Type:	NUK-HP			
Operating pressure max.:				
Boiler contents:				
Steam- and condensate connection		x		
Tube dimension (d _s x s):				
			mm	(4)

A.) Calculation of tubing length to be installed and diameter

Boiler contents				(1)
Safety filling quantity				
= 20% of boiler contents				
(from level Δ_{-1} to Δ_{-2})				
Total filling quantity for start of plant				
(from level Δ_{-2} to Δ_{-3})				
				(5)
				(6)

For the calculation of tubing dimension to be installed (Ø and length) between NUK-HP and the heat exchanger for heating of vegetable oil, the volume expansion of boiler water – relating to the total filling quantity for the start of plant (6) and the operating pressure max. has to be considered.

The volume expansion of boiler water is:

- at heating-up from 20°C / 1 bar to 287°C/ 70 bar excess pressure: 35% (7)
- at heating-up from 20°C/ 1 bar to 304°C/ 90 bar excess pressure: 42% (8)

The contents in the system at operating pressure max. is calculated from the total filling quantity for the start of plant (6) plus the respective volume expansion (7) resp. (8) in regard of (6).

Contents in the system at operating pressure max.				(6)
Total filling quantity for the start of plant (level Δ_{-3})				
Volume expansion (7) resp. (8)				(8)
Contents in the system at operating pressure max.: (level Δ_{-4})				(9)

The contents in the system at operating pressure max. is of importance for the calculation of dimension/ Ø and length of steam- and condensate line.

The steam- and condensate lines are to be effected with the same Ø.

For the determination of Ø a medium operating pressure of saturated steam of 50 bar and a steam velocity of ≤ 10m/s at 50 bar will be taken as basis.

The recommended tube nominal width of GEKAKONUS GMBH for the different boiler of type NUK-HP are listed in a separate table (see page 5).

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For required tubing length between the boiler of type NUK-HP and the heat exchanger for heating of vegetable oil, the contents in the system at operating pressure max. (9) minus the boiler contents (3) is decisive.

Contents in the system at operating pressure max.	=		(9)
minus boiler contents	=		(3)
Tube contents at operating pressure max. (from level V² to V⁴)	=		(10)

Required tubing length, steam- and condensate line are calculated from the contents each meter tubing divided by the tube contents (10).

Total tubing-length form level V² to level V⁴				
Tube contents	(10)	=	Contents per meter tube	meter
Total tubing-length form level V² to level V⁴		=		meter (11)

The total tubing length will be divided to each 50% to the steam – and 50% to the condensate line.

The calculated level V⁴ for reason of safety function of steam generating system shall be min. 0,5 meter below the condensate drain of heat exchanger.

To avoid the likelihood of water entering into the coils of the heating tray, an additional pipe length must be installed (i.e. additional to the 87 meters calculated above). This must be installed such that there is 0.5 meters of empty vertical pipe distance between the water at operational conditions (level 4) and the outlet of the condensate drain of the coil in the heating tray level 5).

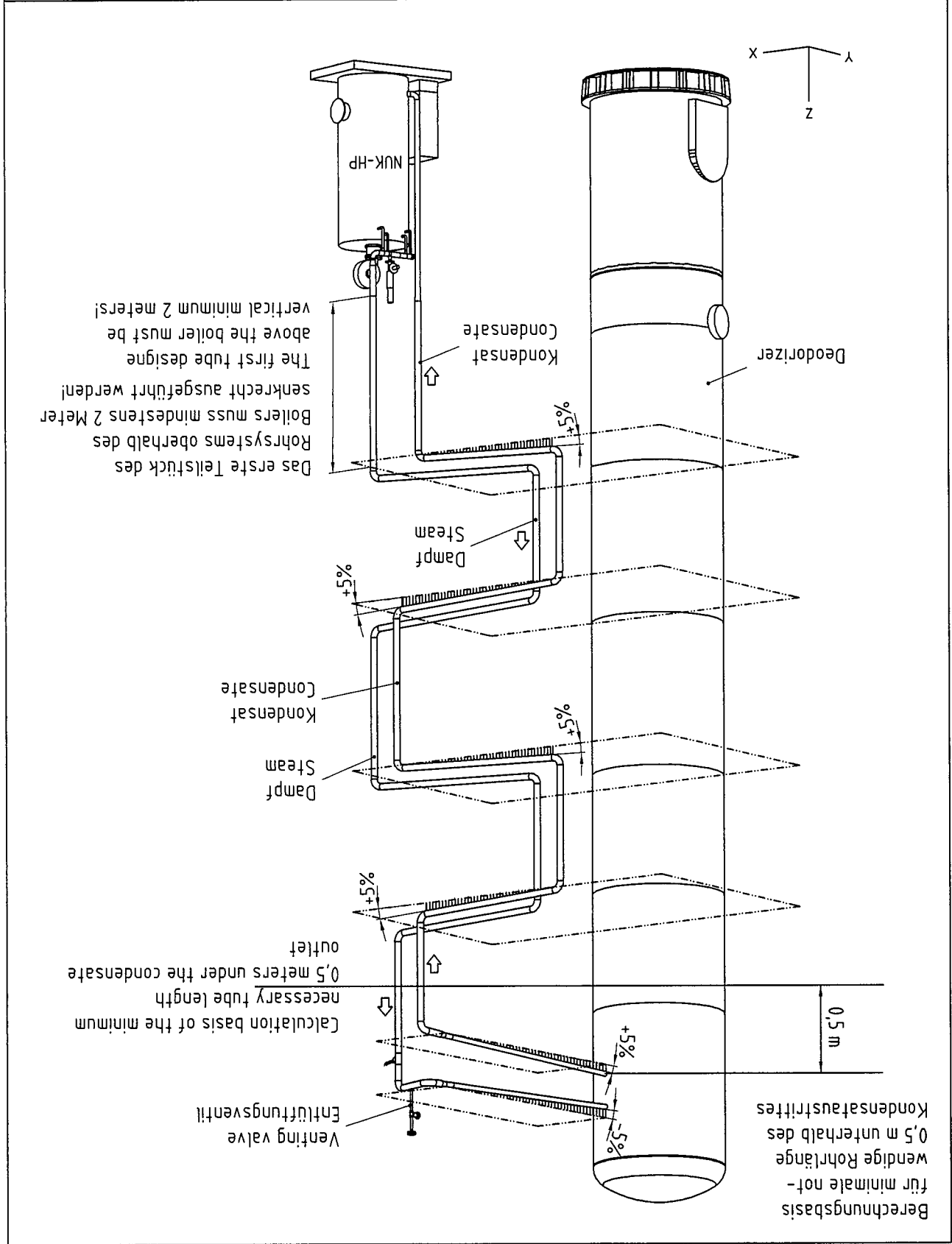
The condensate drain of heat exchanger, level V⁵ shall be arranged min. 2 meter above the boiler type NUK-HP, level V²

if in practice, the required length of piping between level V² and V⁴ can not be installed, or if the required minimum system volume cannot be reached, a water separator (or water buffer vessel) needs to be installed in the system (see Appendix A).

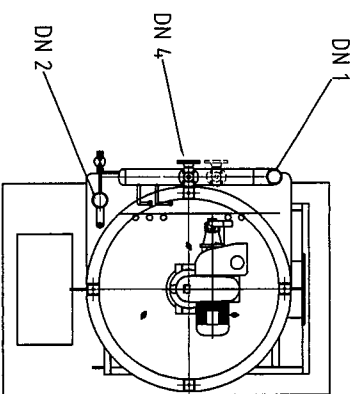
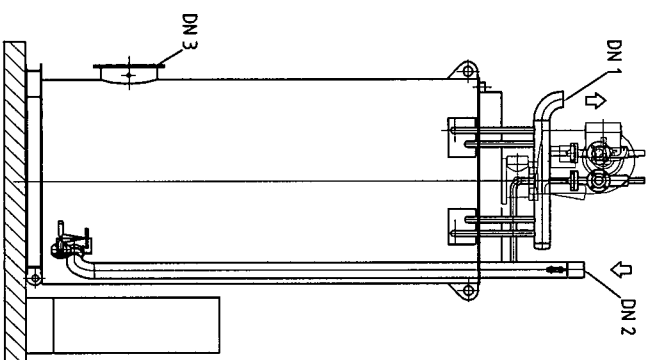
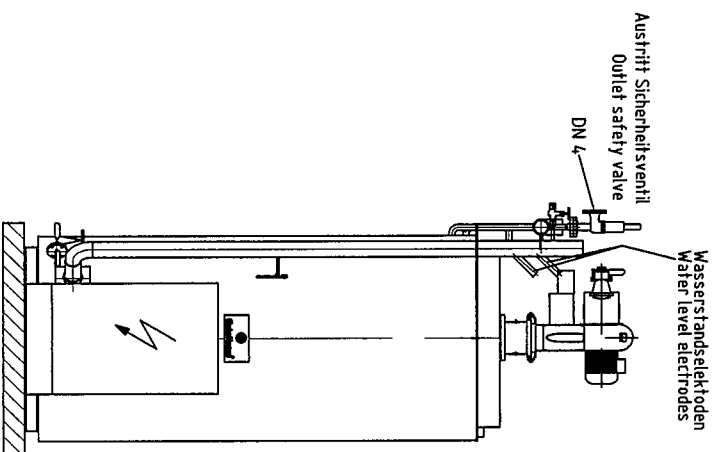
However it must be ensured, that the total tubing length (11) in the area of level V² and V⁴ can be installed.

In case it should be not possible, to find a solution of problem on the strength of above mentioned described procedure, we please ask you, to contact GEKAKONUS GMBH.

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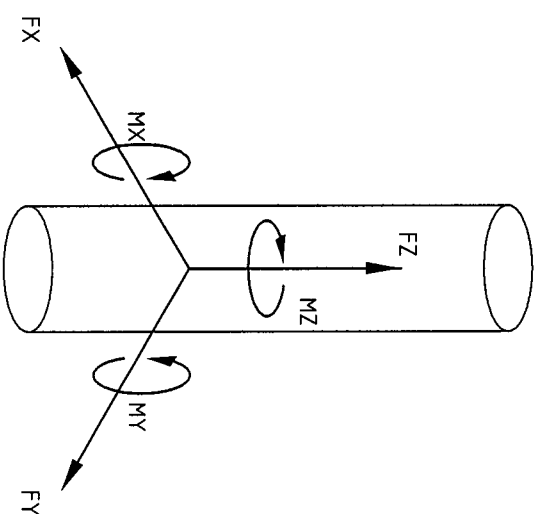


maximum admissible forces / moments on the pipe connection of the boiler



- | | |
|--------|---|
| DN 1 - | Dampf - Schweißende
Stream - weld on end |
| DN 2 - | Kondensat - Schweißende
Condensate - weld on end |
| DN 3 - | Rauchrohranschluss mit Gegenflansch
Flue gas connection with counter flange |
| DN 4 - | DN50-PNk0 ohne Gegenflansch - DIN 2635
DN50-PNk0 without counter flange - DIN 2635 |

Pipe size DN1	$FX=FY=FZ$ [N]	$MX=MY=MZ$ [Nm]	Pipe size DN2	$FX=FY=FZ$ [N]	$MX=MY=MZ$ [Nm]
DN80 (3 inch) 80 / 95 bar	500	100	DN80 (3 inch) 80 / 95 bar	3000	1500
DN100 (4 inch) 80 / 95 bar	800	500	DN100 (4 inch) 80 / 95 bar	5000	2500
DN150 (6 inch) 80 / 95 bar	1000	700	DN150 (6 inch) 80 / 95 bar	6000	3000



Model	15A	15B	15C	15D	15E	15F	15G	15H	15I	15J	15K	15L	15M	15N	15O	15P	15Q	15R	15S	15T	15U	15V	15W	15X	15Y	15Z	15AA	15AB	15AC	15AD	15AE	15AF	15AG	15AH	15AI	15AJ	15AK	15AL	15AM	15AN	15AO	15AP	15AQ	15AR	15AS	15AT	15AU	15AV	15AW	15AX	15AY	15AZ	15BA	15BB	15BC	15BD	15BE	15BF	15BG	15BH	15BI	15BJ	15BK	15BL	15BM	15BN	15BO	15BP	15BQ	15BR	15BS	15BT	15BU	15BV	15BW	15BX	15BY	15BZ	15CA	15CB	15CC	15CD	15CE	15CF	15CG	15CH	15CI	15CJ	15CK	15CL	15CM	15CN	15CO	15CP	15CQ	15CR	15CS	15CT	15CU	15CV	15CW	15CX	15CY	15CZ	15DA	15DB	15DC	15DD	15DE	15DF	15DG	15DH	15DI	15DJ	15DK	15DL	15DM	15DN	15DO	15DP	15DQ	15DR	15DS	15DT	15DU	15DV	15DW	15DX	15DY	15DZ	15EA	15EB	15EC	15ED	15EE	15EF	15EG	15EH	15EI	15EJ	15EK	15EL	15EM	15EN	15EO	15EP	15EQ	15ER	15ES	15ET	15EU	15EV	15EW	15EX	15EY	15EZ	15FA	15FB	15FC	15FD	15FE	15FF	15FG	15FH	15FI	15FJ	15FK	15FL	15FM	15FN	15FO	15FP	15FQ	15FR	15FS	15FT	15FU	15FV	15FW	15FX	15FY	15FZ	15GA	15GB	15GC	15GD	15GE	15GF	15GG	15GH	15GI	15GJ	15GK	15GL	15GM	15GN	15GO	15GP	15GQ	15GR	15GS	15GT	15GU	15GV	15GW	15GX	15GY	15GZ	15HA	15HB	15HC	15HD	15HE	15HF	15HG	15HI	15HJ	15HK	15HL	15HM	15HN	15HO	15HP	15HQ	15HR	15HS	15HT	15HU	15HV	15HW	15HX	15HY	15HZ	15IA	15IB	15IC	15ID	15IE	15IF	15IG	15IH	15II	15IJ	15IK	15IL	15IM	15IN	15IO	15IP	15IQ	15IR	15IS	15IT	15IU	15IV	15IW	15IX	15IY	15IZ	15JA	15JB	15JC	15JD	15JE	15JF	15JG	15JH	15JI	15JJ	15JK	15JL	15JM	15JN	15JO	15JP	15JQ	15JR	15JS	15JT	15JU	15JV	15JW	15JX	15JY	15JZ	15KA	15KB	15KC	15KD	15KE	15KF	15KG	15KH	15KI	15KJ	15KL	15KM	15KN	15KO	15KP	15KQ	15KR	15KS	15KT	15KU	15KV	15KW	15KX	15KY	15KZ	15LA	15LB	15LC	15LD	15LE	15LF	15LG	15LH	15LI	15LJ	15LK	15LL	15LM	15LN	15LO	15LP	15LQ	15LR	15LS	15LT	15LU	15LV	15LW	15LX	15LY	15LZ	15MA	15MB	15MC	15MD	15ME	15MF	15MG	15MH	15MI	15MJ	15MK	15ML	15MN	15MO	15MP	15MQ	15MR	15MS	15MT	15MU	15MV	15MW	15MX	15MY	15MZ	15NA	15NB	15NC	15ND	15NE	15NF	15NG	15NH	15NI	15NJ	15NK	15NL	15NM	15NN	15NO	15NP	15NQ	15NR	15NS	15NT	15NU	15NV	15NW	15NX	15NY	15NZ	15OA	15OB	15OC	15OD	15OE	15OF	15OG	15OH	15OI	15OJ	15OK	15OL	15OM	15ON	15OO	15OP	15OQ	15OR	15OS	15OT	15OU	15OV	15OW	15OX	15OY	15OZ	15PA	15PB	15PC	15PD	15PE	15PF	15PG	15PH	15PI	15PJ	15PK	15PL	15PM	15PN	15PO	15PP	15PQ	15PR	15PS	15PT	15PU	15PV	15PW	15PX	15PY	15PZ	15QA	15QB	15QC	15QD	15QE	15QF	15QG	15QH	15QI	15QJ	15QK	15QL	15QM	15QN	15QO	15QP	15QQ	15QR	15QS	15QT	15QU	15QV	15QW	15QX	15QY	15QZ	15RA	15RB	15RC	15RD	15RE	15RF	15RG	15RH	15RI	15RJ	15RK	15RL	15RM	15RN	15RO	15RP	15RQ	15RR	15RS	15RT	15RU	15RV	15RW	15RX	15RY	15RZ	15SA	15SB	15SC	15SD	15SE	15SF	15SG	15SH	15SI	15SJ	15SK	15SL
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Appendix H Operation instructions for pump

Manufacturer:

Speck

Type:

P 11/15-150

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The GEKAKONUS GMBH NUK-HP is working in a hermetically steam-/condensate system without automatic levelling feedwater supply. The provided feed water refilling pump is only used for charging and re-charging the plant. The re-charging is manually operated and required during a low-water fault/breaking, caused by water shortage. The refilling pump can also be used for the pressure test of the system with cold water.

ATTENTION !

The charging and re- charging of the plant must only be done during pressureless operating conditions. Based on a regular installation, a simultaneous operation of feed water pump and burner is not possible. The electrical equipment for the operation of feed water filling pump are reciprocally locked in the switchboard. The electrical connection of the pump happens via a socket, installed at the outer side of the switchboard. The power supply of the socket happens via a socket at the switchboard.

The pump is mounted - ready for operation - on a base frame complete with an electrical motor, an electrical cable with plug, a cone belt and a belt guard.

The required pipe connections to the GEKAKONUS GMBH NUK-HP are in flexible design with a high-pressure hose.

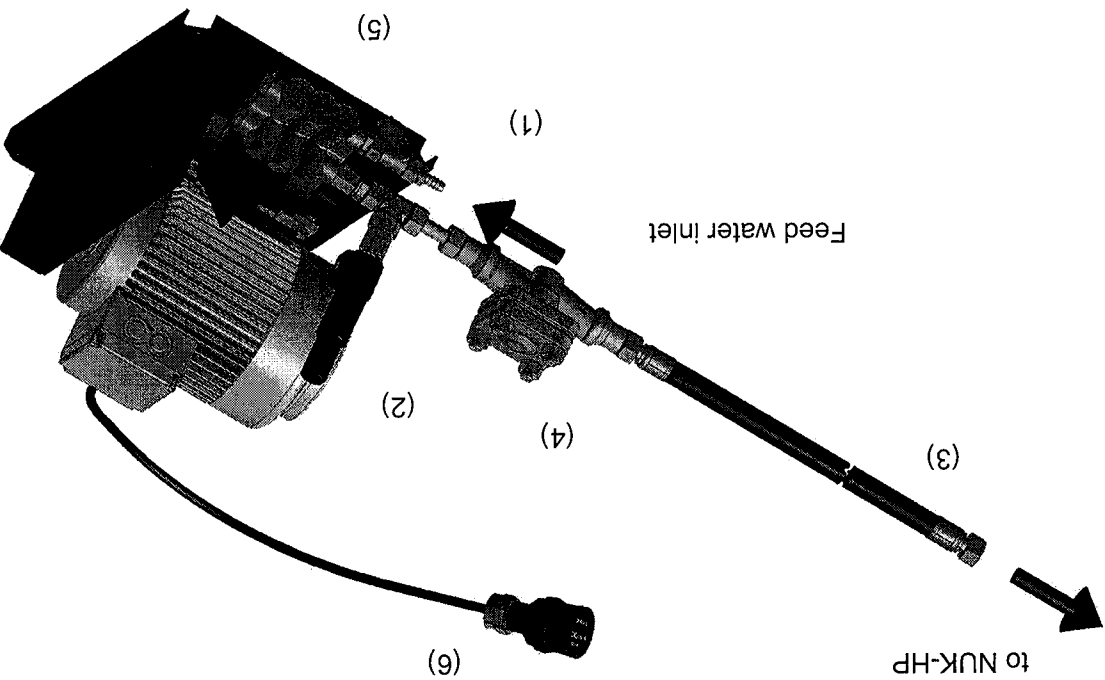
For safety reasons between high-pressure hoses and the filling/drain valve of the NUK-HP a non-return valve is installed. This prevents at burst of hoses a dangerous discharge of cold/hot water or steam.

After the charging resp. re-charging the electrical and the water side connection to the GEKAKONUS GMBH NUK-HP must be released, the filling-opening must be closed with the delivered fittings and the filling/drain valve must be closed too.

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STANDARD INSTRUMENTATION:

- 1 pc. strainer R 1/2" (1)
- 1 pc. safety valve for feed water R1/2", 140 bar (2)
- 1 pc. high pressure hose with couplings R 1/2" (3)
- 1 pc non return valve (4)
- 1 pc. plunger pump type P11/15-150 (5)
- 1 pc. electrical connection to NUK-HP switchboard (6)



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Technical Specifications

Type	P 11/15-150
Power consumption	2,2 kW
Pressure max.	150 bar
RPM max.	1420 min ⁻¹
Output max.	14,7 l/min.
Water-temp. max.	70°C
Plunger dia.	18,0 mm
Stroke	14,1 mm
Weight approx.	6,3 kg
NPSH Required	5,3 m.l.c.

Required NPSH refers to water : Specific weight 1 kg/dm³, viscosity 1 cP at max. permissible revolutions.

Operation and Maintenance

Check oil level prior to starting and ensure trouble-free water supply. Oil : Use only 0.26 litres of ISO VG 220 (e.g. Aral Degol BG 220) or SAE 90 gear oil.

Initial change after 50 operating hours and then every 500 operating hours, after 6 months operation in any case.

ATTENTION!

When operating in damp places or with high temperature fluctuations. Oil must be changed immediately, should condensate (frothy oil) occur in the gear box. Max. input pressure 10 bar. Keep NPSH under control.

Safety and Warranty rules

There is a 6 months warranty in accordance with VDMA guidelines.

Pump operation without safety valve as well as any excess in temperature or speed limits automatically voids the warranty. The safety valve must be regulated in accordance with the guidelines for liquid spraying units so that the admissible operating pressure can not be exceeded by more than 10 %.

When the pump is in operation, the open shaft end must be covered up by shaft protector (17), the driven shaft side and coupling by a contact protector.

Pressure in discharge line and in pump must be at zero before any maintenance to the pump takes place. Close up suction line. Disconnect fuses to ensure that the driving motor does not get switched on accidentally.

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ATTENTION!

Make sure that all parts on the pressure side of the unit are vented and refilled, with pressure at zero, before starting the pump. In order to prevent air, or an air/water-mixture being absorbed and to prevent cavitation occurring, the pump NPSHR, positive suction head and water temperature must be kept under control. Cavitation and / or compression of gases lead to uncontrollable pressure-kicks which can ruin pump and unit parts and also be dangerous to the operator or anyone standing nearby.

Plunger Pumps are suitable for pumping clean water and other non-aggressive or abrasive media with a specific weight similar to water. On no account may inflammable or explosive media be pumped.

MAINTENANCE

To check valves

Suction Valve: Screw out plugs (41). Take out suction valve adapter (39) together with suction valve. Push valve parts out of suction valve adapter using a soft tool. Check and replace worn parts.

Check O-rings (38,40,42) and replace as necessary. Tighten plugs (41, 43) to 70 NM.

To check Seals and Plunger Pipe

Screw out plugs (41). Loosen nuts (46) and remove valve casing from plungers, pulling it out to the front. Take out suction valve adapter (39), tension spring (33) and seal-unit (30,31,32). Check surfaces of plunger pipes as damaged surfaces cause fast wear to the seals.

When replacing V-sleeves (31), grease new seals with special grease from pump manufacturer before installing. Check O-rings (40,42) and replace as necessary.

ATTENTION !

Care must be taken that no glue gets between the plunger pipe (24A) and centring on plunger (22). The plunger pipe should not be strained by eccentric tightening of tension screw nor through dirt or damage to the front surface of the plunger as this could cause the plunger pipe to break.

Install tension spring (33) and suction valve adapter (39), then tighten plug (41) to 70 NM.

Fix valve case by tightening nuts (46) evenly to 47,5 NM.

To check drip return

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After removing the valve casing, the V-sleeves (50) in the intermediate casing (48) can be examined and replaced if necessary. Particular care must be taken that the two bores in the valve casing are free of lime deposits so as not to prevent the drip return.

To dismantle gear

Drain oil after dismantling the valve casing (29) and intermediate casing (48), then screw off crankcase cover (3) and bearing cover (12). Whilst turning slightly, hit out the crankshaft to one side with a rubber hammer. Do not force the cranks on the shaft when pulling through the connecting rod. If necessary, either press out second bearing in crankcase or hit it out carefully with a soft tool.

ATTENTION !

Connecting rods are marked for identification. Con rod is to be reinstalled in the same position on shaft journals. Do not bend the con rod shanks. Check shaft and con rod surfaces, shaft seals (26), crosshead + plunger (22) and ball bearings.

To reassemble

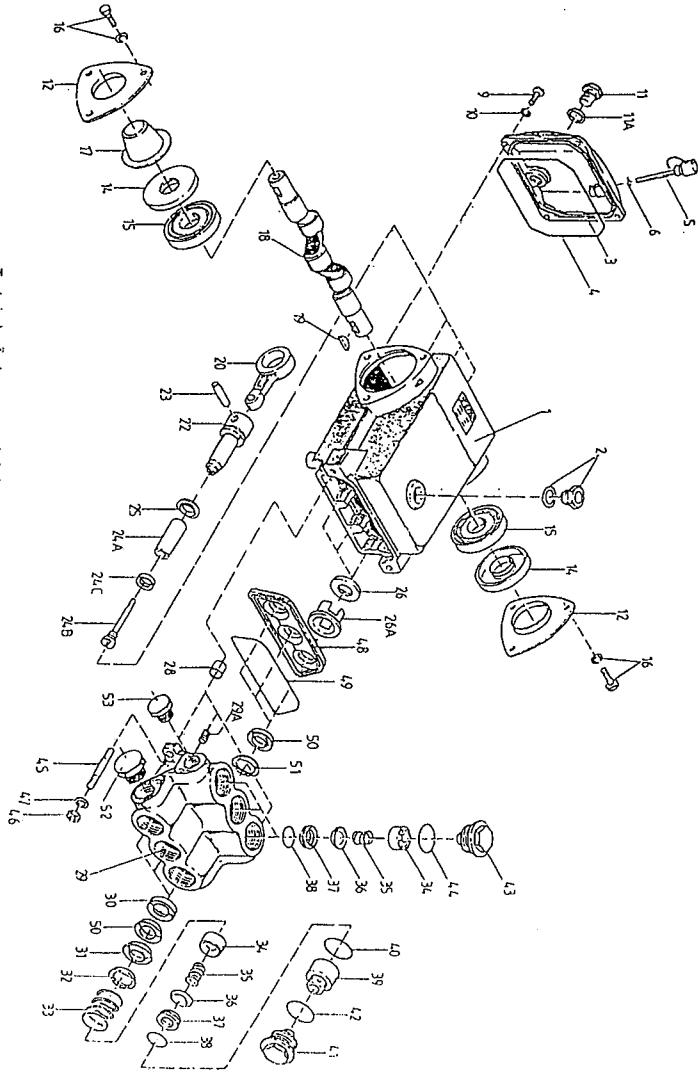
Using a soft tool, press one ball bearing in till it reaches the edge of the bearing hole. Push shaft with the other bearing in carefully through the opposite bearing hole and fit in carefully through the con rods. Press on bearings on both sides. Carefully thread shaft seal onto shaft and press into casing. Fit on bearing cover (12).

ATTENTION !

After assembly has been completed, the shaft should turn easily with very little clearance.

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Technische Änderungen vorbehalten
 Subject to change
 P11 D

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Spare parts list P 11 / 15-150

Item no.	No. Off	Code no.	Description
1	1	01.0269	Crankcase
2	1	00.2373	Vent / Filler Plug Assy
3	1	03.0121	Crankcase Cover
4	1	06.0050	O-Ring for 3
5	1	00.0901	Oil-Dipstick Assy
6	1	06.053	O-Ring for 5
9	4	21.0069	Cylinder Screw
10	4	07.0595	Spring Ring
11	1	07.0487	Oil Drain Plug
11A	1	06.0054	Gasket
12	2	03.0159	Bearing Cover
14	2	06.0055	Radial Shaft Seal
15	2	05.0076	Grooved Ball Bearing
16	6	21.0028	Hexagon Screw
17	1	07.2530	Shaft Protector
18	1	11.0351	Crankshaft (P11/10-100)
18	1	11.0107	Crankshaft (P11/13-100)
18	1	11.0255	Crankshaft (P11/15-150)
19	1	07.0596	Woodruff Key
20	3	16.0018	Connecting Rod
22	3	00.0597	Crosshead Assy
23	3	11.0066	Crosshead Pin
24A	3	11.0115	Plunger Pipe
24B	3	21.0351	Tension Screw
24C	3	06.0306	Copper Ring
25	3	07.2513	Oil Scraper
26	3	06.0064	Radial Shaft Seal
26A	3	072964	Spacer Sleeve.
28	2	07.0558	Centring Sleeve
29	1	01.0425	Valve Casing
30	3	07.0879	Pressure Ring

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When ordering please state Code no., Pump Model and Pump Serial no.

31*	3	06.1133	Sleeve
32	3	07.0880	Support Ring
33	3	07.0881	Pressure Spring
34**	6	07.3087	Spring Tension Disc
35**	6	07.1941	Valve Spring
36**	6	07.1004	Valve Plate
37**	6	07.0292	Valve Seat
38**	6	06.0067	O- Ring
39	3	07.0882	Suction Valve Adapter
40*	3	06.0271	O- Ring
41	3	07.0878	Plug M 28x1.5
42*	3	06.0272	O-Ring
43	3	00.1514	Plug G 3/4
44*	3	06.0496	O- Ring
45	2	21.0078	Stud Bolt
46	2	07.2398	Hexagon Nut
47	2	07.2706	Disc
48	1	01.0249	Intermediate Casing
49*	1	06.0869	O- Ring
50*	3	06.1057	Sleeve
51*	3	07.0941	Support Ring
52	1	07.0705	Plug G 1/2
53	1	07.0608	Plug G 3/8
	1	00.1963	Crankcase Assy
		P 11/15 (1-28/45-47)	
	1	00.0785	Pump Head Assy
*	1	14.0494	Seal Repair Kit
**	1	14.0414	Valve Repair Kit

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Appendix I

Level control electrode NRG 19-50 and level switch NRS 1-50

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1.0 LEVEL CONTROL ELECTRODE NRG 19-50

1.1 FUNCTION

The level electrodes NRG 19-50 are used in conjunction with level switch NRS 1-50 as water level limiters for steam boiler plants. Water level limiters switch off the heating when the water level falls below the set minimum level (low water).

1.2 DIRECTIVES AND STANDARDS

Pressure Equipment Directive (PED) 97/23/EC (only in conjunction with NRS 1-50)
EC type approved according to EN 12952-11, EN 12953-09
These directives state, among other things, the requirements made on limiting systems and equipment for steam boiler plants.

Functional Safety according to IEC 61508 (only in conjunction with NRS 1-50)
This standard describes the functional safety of safety-related electrical / electronic / programmable electronic systems.

VdTUV Bulletin "Wasserstand 100" (= Water Level 100)
The NRG 19-50 in conjunction with NRS 1-50 are type approved according to the VdTUV Bulletin "Water Level 100".

1.3 TECHNICAL DATA

Type: NRG 19-50
Maximum service pressure: 100 bar (psig)
Saturation temperature: 311°C
Nominal pressure: PN 160
Connections: G 3/4"
Type approval-no.: TÜV.SWB.XX-XXX

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1.4 INSTALLATION

GESTRA level- control electrodes are high-quality measuring probes and should be handled with care. Avoid subjecting the electrode tip to shocks or bending as this might cause damage to the electrode seal.

Step 1

1. Screw electrode tip (4) into measuring electrode (1) (see Appendix I, page 4).
2. Carefully determine required measuring length of electrode
3. Mark length of electrode tip (4)
4. Unscrew electrode tip (4) from measuring electrode (1) and cut tip
5. After visual inspection screw electrode tip (4) into measuring electrode (1). Slide spring (3) along electrode tip, so that its end completely enters into the small bore (2)

Step 2

1. Check the seating surface of the thread or flange provided on the vessel or boiler standpipe
 2. Place ring joint (5) onto seating surface (6) of electrode
 3. Apply a light smear of silicone grease (e.g. WMINIX 2150) to electrode thread (7)
 4. Screw level electrode into threads provided on the boiler and tighten with a 41 mm open-end spanner. The torque required is 160 Nm when cold
- Piece of advice:

- The seating surface of the thread or flange provided on the vessel of the boiler standpipe must be accurately machined
- Do not bend the electrode tip when mounting
- Do not lag the electrode body situated above the hexagonal part

1.5 INSPECTION AND TESTING

The electrode is of self-testing type, if a test failed this mail function will be indicated. For more details see next chapter (Level switch NRS 1-50)

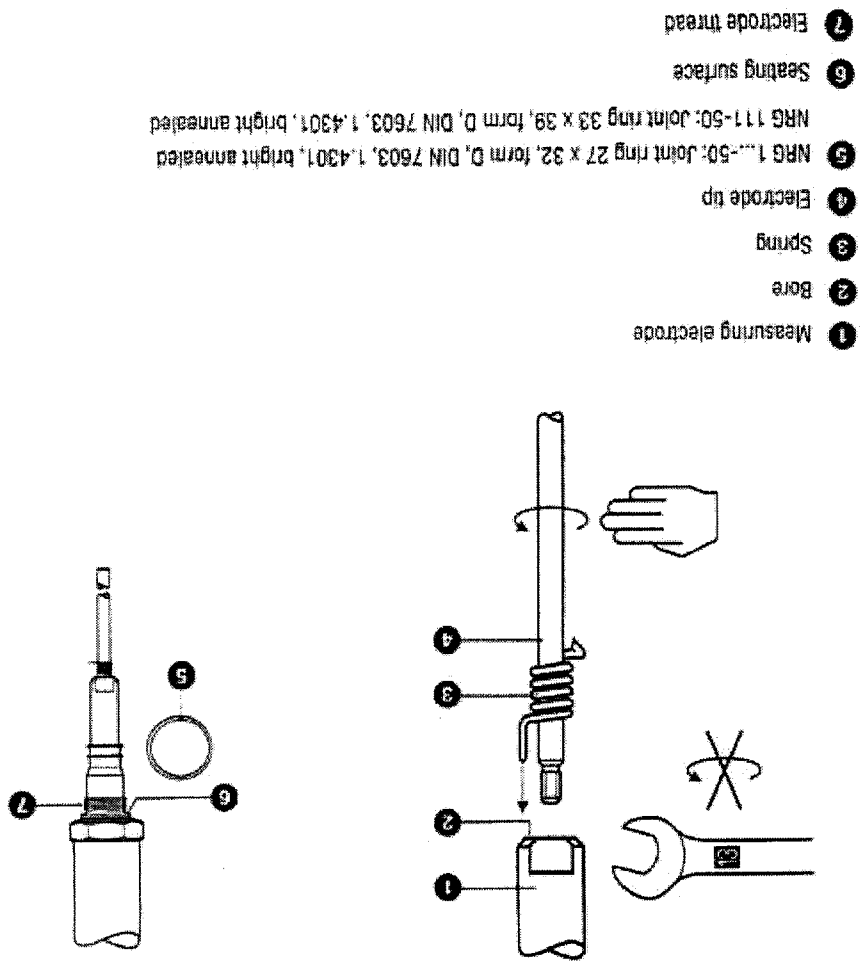
1.6 OPERATION

During the normal operation the green LED "Operation" indicates a normal function of the electrode. If the water is too low or there is a mail function of the electrode the red alarm light is on. (see description level switch NRS 1-50)

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1.7 MAINTENANCE

No user serviceable parts are contained within the equipment. Only the manufacturer must perform all repairs. Misuse or any attempted modification of the equipment will lead to considerable safety risks.



- 1 Measuring electrode
- 2 Bore
- 3 Spring
- 4 Electrode tip
- 5 NRG 1...-50: Joint ring 27 x 32, form D, DIN 7603, 1.4301, bright annealed
- NRG 111-50: Joint ring 33 x 39, form D, DIN 7603, 1.4301, bright annealed
- 6 Seating surface
- 7 Electrode thread

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2.0 LEVEL SWITCH NRS 1 - 50

2.1 FUNCTION

The level switch NRS 1-50 in conjunction with the NRG 19-50 is self-monitoring water level limiter with periodical self-test. This combination monitors the lowest water level in the boiler and switches off the heating when the water level falls below the set minimum level.

2.2 TECHNICAL DATA

Safety Integrity Level	Classification of the Safety Integrity Level acc. to IEC 61508
SIL	SIL 3 (only in conjunction with NRG 19-50)
Safe Failure Fraction	Percentage of failures without the potential to put the safety-related system into a dangerous state
SFF	97,80% (in combination with 2 electrodes NRG 19-50
Type- approval no.:	TÜV-SWB.XX-422
Output safety circuit:	Two volt free make contacts, 24 V DC – 250 V AC, max. 6A (resistive/inductive), contact material AgNi.
Signal output:	2 volt-free outputs for instantaneous external signalling, 24-230 V AC/DC, max. 100 mA
Response sensitivity:	> 10...<10000 µS/cm at water temperature of 25°C
Indicators and adjusters:	2 buttons for test and diagnosis, 4 red/green LEDs for indicating the operation mode and alarm.
Mains supply:	6 red LEDs for fault diagnosis, 2 two-pole code switches for setting the number of electrodes.
Protection:	See Wiring diagram
Perm. ambient temperature:	Housing: IP 40 to EN 60529
	Terminal strip: IP 20 to 60529
	0...55°C

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2.3 INSTALLATION

On supporting rail (with mounting clip)

- Snap the level switch onto the supporting rail

Wiring should be carried out in accordance with the wiring diagram for the project (Appendix J).

Connect screens of electrode wires only to terminals 5 and 13.

At operation with two level electrodes NRG 19-50 have to be the code switches (4) and (5) in position "OFF"!

2.4 TESTING

A low-level alarm can be simulated by pushing the button "1" or "2" whilst the electrode tip is submerged. Red LED of level electrode is flashing until the response delay has expired. After this the red LED has to illuminate. The test is finished.

2.5 OPERATION

When the green LEDs illuminate during the normal operation then the water level is enough. The red LEDs illuminate when the water level falls below the minimum level or there is a malfunction of the electrode.

2.6 MAINTENANCE

No user serviceable parts are contained within the equipment. Only the manufacturer must perform all repairs. Misuse or attempted modification of the equipment will endanger plant safety.

2.7 TROUBLESHOOTING

Before carrying out the fault diagnosis please check:

Mains supply:

Is the level switch supplied with the mains voltage specified on the name plate?

Wiring:

Is the wiring in accordance with the wiring diagram and the relevant schematic representation of arrangement?

Configuration:

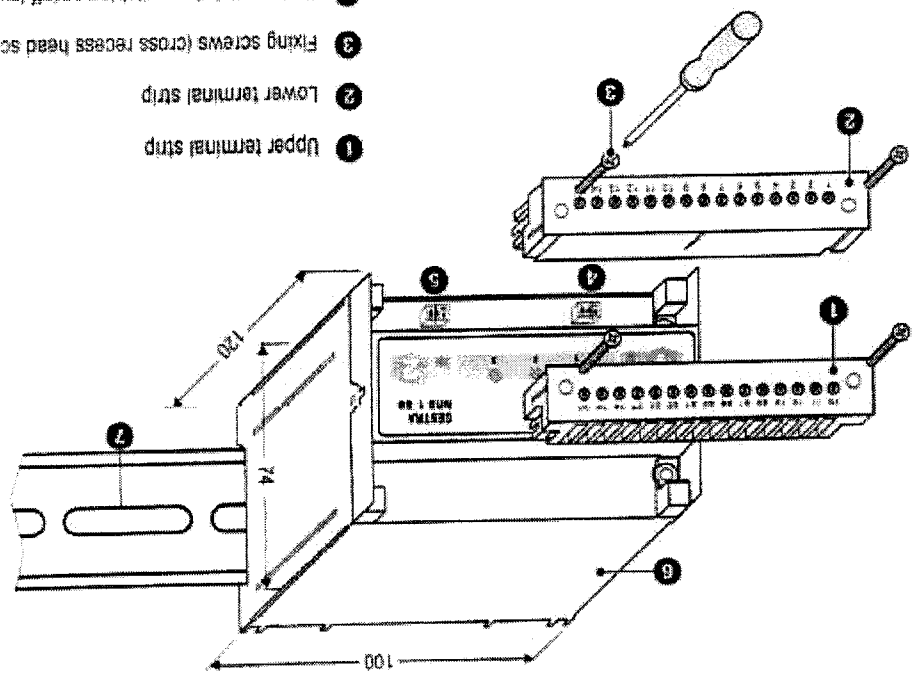
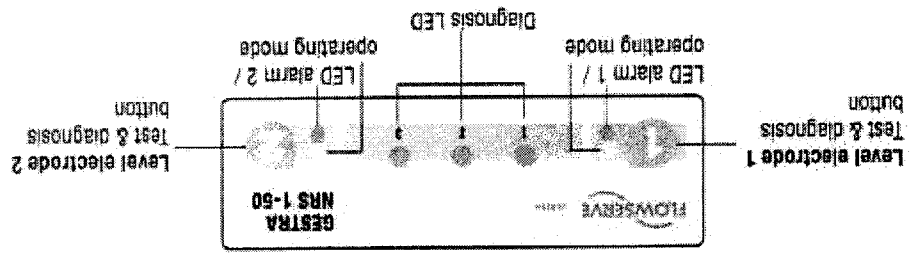
Are the code switch settings (4) and (5) correct for the number of level electrodes used?

 Gekakonus®		OPERATING INSTRUCTIONS NATURAL CIRCULATION STEAM BOILER TYPE NUK-HP	
LEVEL CONTROL ELECTRODE NRG 19-50 AND LEVEL SWITCH NRS 1-50			
Revision 001	Date 23.03.11	Chapter Appendix I	Page 7 of 8

Diagnosis			
Display 1 and activity	Display 2	Fault	Remedy
LED alarm 1 and diagnosis LED 1 illuminated. Press and hold down key 1.	Diagnosis LED1 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring, faulty measuring voltage.	- check wiring, - measure electrode voltage - clean and if necessary exchange level electrode switch
	Diagnosis LED2 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring.	- exchange level electrode switch
	Diagnosis LED3 flashing.	Interference voltage causing malfunction, boiler earth without PE.	Provide screen and earthing, connect boiler with PE.
LED alarm 2 and diagnosis LED 2 illuminated. Press and hold down key 2.	Diagnosis LED1 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring, faulty measuring voltage.	- check wiring, - measure electrode voltage - clean and if necessary exchange level electrode switch
	Diagnosis LED2 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring.	- exchange level electrode switch
	Diagnosis LED3 flashing.	Interference voltage causing malfunction, boiler earth without PE.	Provide screen and earthing, connect boiler with PE.
LED alarm 1 and 2 and diagnosis LED 3 illuminated. Press and hold down key 1 or 2.	Diagnosis LED1 flashing.	Malfunction in processor, stand-by fault.	Observe operating instructions for the logic unit SRL. Replace level switch.
	Diagnosis LED2 flashing.	Internal voltage fault.	Replace level switch.
	Diagnosis LED3 flashing.	Malfunction in relay.	

Once the fault is eliminated, the level switch returns to normal operation.
After elimination of the fault switch off the mains voltage and switch it on again after approx. 5 sec.

 Gekakonus®		OPERATING INSTRUCTIONS NATURAL CIRCULATION STEAM BOILER TYPE NUK-HP	
LEVEL CONTROL ELECTRODE NRG 19-50 AND LEVEL SWITCH NRS 1-50			
Revision 001	Date 23.03.11	Chapter I Appendix I	Page 8 of 8



- 1 Upper terminal strip
- 2 Lower terminal strip
- 3 Fixing screws (cross recess head screws M3)
- 4 Code switch for switching on/off level electrode 1 / 2
- 5 Code switch for switching on/off level electrode 1 / 2
- 6 Enclosure
- 7 Supporting rail type TH 35, EN 60715

 Gekakonus®		OPERATING INSTRUCTIONS NATURAL CIRCULATION STEAM BOILER TYPE NUK-HP	
WIRING DIAGRAM			
Revision 000	Date 01.04.08	Chapter J Appendix J	

Appendix J
Wiring diagram
Wiring diagram
List of parts

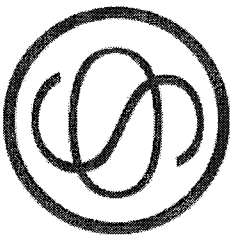
designed with ELCAD-AUCOTEC (R) 7.11.0

SES-Schaltanlagen GmbH
Etogesstrasse 42
D-76275 Ettlingen

SES-Schaltanlagen GmbH
Kruppstrasse 8A
D-76344 Eggenstein

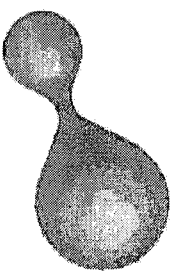
Tel.: +49 (0) 721 / 9419424 / 25
Fax: +49 (0) 721 9419426
www.ses-schaltanlagen.de
info@ses-schaltanlagen.de

power supply	:	3x380V/50Hz
control voltage	:	220V
signal voltage	:	220V / 24V
standard	:	CENELEC HD 384 / EN 50156-1 (VDE 0100) (VDE 0116)
system of protection	:	IP54



Gekakonius®

Internet: www.gekakonus.net
E-mail: info@gekakonus.net
Phone +49 (0) 721 / 94374-0
Fax +49 (0) 721 / 9437444



Ipicco

www.ipicco.com

wiring diagram

customer : 
part of plant : NUK-HP-930
drawing no. : 10-9851
cabinet : 1000x1400x400mm WHD

iteration	date	name	norm	date	proc.	rev.	substitute through	substitute for	origin	wiring diagram SES-18-3898	Sheet 0
				05.11.2018	E.H.					NUK-HP-930 930ar cover 1	54 Bl.

4

Der Schaltplan besteht aus Einzelblättern welche durch Blatt-Nr. und Blatt-Anzahl gekennzeichnet sind

The wiring diagram is composed of separate sheets, marked with sheet-number and total number of the sheets.

 WARNING

Installation, operation and maintenance procedures performed by unqualified personnel may result in improper adjustment and unsafe operation. Either condition may result in equipment damage or personnel injury. Only qualified and specific trained personnel are allowed to carry out the installation, commissioning, operation and maintain the series boiler/heater and the boiler/heat exchanger equipment. The boiler/heat exchanger equipment is tested of logic function prior the boiler/heat exchanger leave the Gaskinox manufacturing shop. All actions on the boiler/heater and boiler/heat exchanger equipment done by the own and/or foreign staffs is on the risk and responsibility of the buyer and/or operator of the boiler/heater.

Scope of Manual

The boiler/heater instruction manual provides installation, adjustment, maintenance instructions and controller configuration data. Complete instructions of the series boiler/heat and boiler/heat equipment found in the instruction manuals.

WELD CAUTION

Switch off the main switch on the switchboard and interrupt the protection earth (PE) cable before beginning of any welding on the boiler/heater piping system. Do not connect the ground connection of the welding set on any part of the boiler/heater during the welding on the boiler/heater piping system. The electrical boiler/heater system is endangered and can be destroyed.

Connect the protection earth (PE) cable again when the welding procedure is concluded.



D-75344 Eggman-Lee

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[illegible]

wiring diagram SES-18-3898

NUK-HF
930
95bar

10-9851

sheet 0A	
54 RI	

1		2		3		4		5		6		7		8	
boiler		type		sheet		name of sheet				date		special memo			
NUK-HP		1500		0		cover 1				22.05.2018					
NUK-HP		1500		0A		cover 2				22.05.2018					
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NUK-HP		1500		0C		table of contents				22.05.2018					
NUK-HP		1500		A1		part list				22.05.2018					
NUK-HP		1500		A2		part list				22.05.2018					
NUK-HP		1500		A3		part list				22.05.2018					
NUK-HP		1500		B		arrangement drawing mounting plate				22.05.2018					
NUK-HP		1500		C		arrangement drawing panel door				22.05.2018					
NUK-HP		1500		1		feeding				22.05.2018					
NUK-HP		1500		2		emergency stop				22.05.2018					
NUK-HP		1500		3		primary circuit				22.05.2018					
NUK-HP		1500		4		primary circuit				22.05.2018					
NUK-HP		1500		5		pump control				22.05.2018					
NUK-HP		1500		6		burner motor control				22.05.2018					
NUK-HP		1500		7		cooling air fan,preheater control				22.05.2018					
NUK-HP		1500		8		SPS-LOGO control				22.05.2018					
NUK-HP		1500		9		safety circuit				22.05.2018					
NUK-HP		1500		10		safety circuit				22.05.2018					
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NUK-HP		1500		39		control voltage 24VDC				22.05.2018					
NUK-HP		1500		40		spare				22.05.2018					

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	NUK-HP	1500	42	connection diagram -X2, -X3, -X4, -X6, XL, -XN		22.05.2018		
	NUK-HP	1500	43	connection diagram -XS1, -XS2, -XS3		22.05.2018		
	NUK-HP	1500	44	connection diagram -XS4, -XS5, -XS6		22.05.2018		
	NUK-HP	1500	45	connection diagram -XS7, -XS8, -XS9, -XS10		22.05.2018		
B								
C								
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E								
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F								
writing diagram SES-18-3898								
NUK-HP								
930								
table of contents								
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54 Bl.								
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comments :

comments :

[illegible]

part list

equipment identifier	item no.	class	manufacturer	Typ	comment	refer to
-K26	0040003054	K	FI	55.32 9 024 00 40	auxiliary relay	/2.E6
-K53	0041000030	K	KM	DILM9-10	main contactor	/5.E3
-K55	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/5.E5
-K57	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/5.E7
-K72	0041000030	K	KM	DILM9-10	main contactor	/7.E2
-K73	0040003056	K	FI	55.34 8 230 50 40	auxiliary relay	/7.E3
-K75	0041000034	K	KM	DILM17-10	main contactor	/7.E5
-K77	0041000034	K	KM	DILM17-10	main contactor	/7.E7
-K97	0041000003	K	KM	DILA-22	auxiliary relay	/9.E7
-K98	0030000022	K	ABB	NF22E-13	auxiliary relay	/9.E8
-K103	0041000003	K	KM	DILA-22	auxiliary relay	/10.D3
-K104	0030000022	K	ABB	NF22E-13	auxiliary relay	/10.D4
-K127	0041000003	K	KM	DILA-22	auxiliary relay	/12.E7
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-K147	0041000002	K	KM	DILA-31	auxiliary relay	/14.E7
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-K152	0030000002	K	ABB	N 31E	auxiliary relay	/15.D2
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-K176	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/17.D6
-K197	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/19.E7
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-K262	0040003056	K	FI	55.34 8 230 50 40	auxiliary relay	/26.E2
-K263	0040003056	K	FI	55.34 8 230 50 40	auxiliary relay	/26.E3
-K271	0030000022	K	ABB	NF22E-13	auxiliary relay	/27.D1
-K272	0041000056	K	KM	ETR2-12	timing relay	/27.D2
-K273	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/27.D2
-K274	0050000300	K	SI	3RT2023-1AP	main contactor	/27.D3
-K274	0050001004	K	SI	3RT2926-2PA01	timing unit cut in delayed	/27.D3
-K275	0041000030	K	KM	DILM9-10	main contactor	/27.D4
-K286	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/28.E5
-K322	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/32.D2
-K323	0030000021	K	ABB	NF31E-13	auxiliary relay	/32.D3
-K324	0030000021	K	ABB	NF31E-13	auxiliary relay	/32.D4

comments :

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comments :

comments :

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05.11.2018	E.H.	

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origin

Gelekonius

writing diagram SFS-18-3898

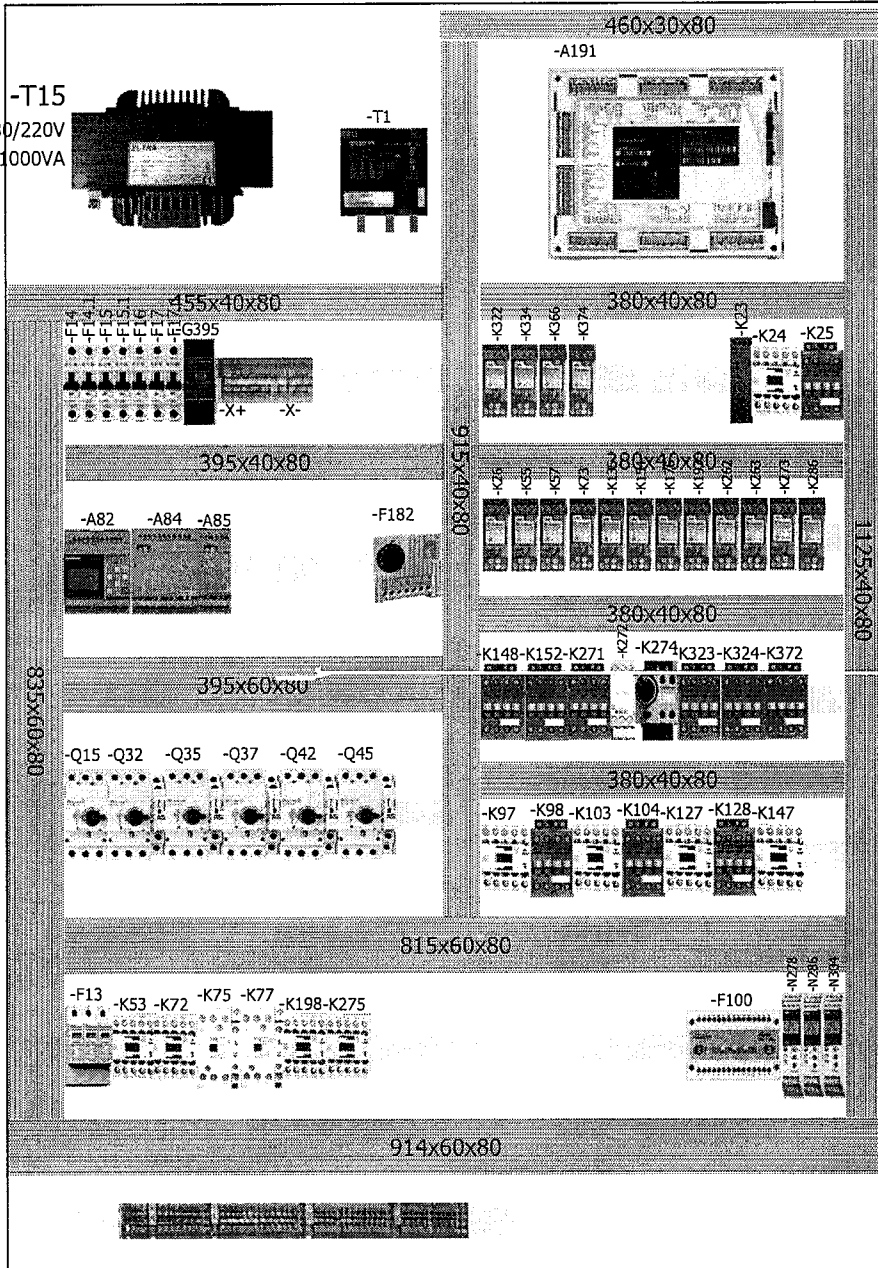
arrangement drawing mounting plate

scale : 1:5

measure unit : mm

sheet B

54 Bl.



alteration	date	name	nom
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date	proc.	rev.
05.11.2018	E.H.	

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origin

writing diagram SES-18-3898

arrangement drawing panel door

10-9851

sheet C

54 Bl.

scale : 1:5

measure unit : mm

Q11

NLT100

A87=SIEMENS-Logo TD

A213=combustion manager indication-/operation unit

N120=pressure controller/temperature controller

H177=hoooter

H76=preheater in operation

H287=oil temperature too low

H288=oil temperature too high

H56=motor protection switch failure

S366/H368= cascade controlled

S167=hoooter off

SS2/H52=filling pump start/stop

H27=control voltage on

H275=oil pressure pump in operation

H263=burner firing gas

H264=burner firing fuel oil

S302/H303= remote set point pressure

off on

H74=cooling air fan fault

H377=burner on external start

H378=burner on external stop

S72/H72=cooling air fan

S205=burner on

S207=burner sequence

1=automatic operation

2=partial load,only

3=servomotor stop

4=full load,only

S24/H28=reset emergency-stop

S152/H153=reset boiler

S196/H196=reset combustion manager

S285= preselection switch

1=gas

2=high speed diesel

3=medium fuel oil

Q11=main switch

S22=emergency-stop

H55=extem fuel oil pump HSD

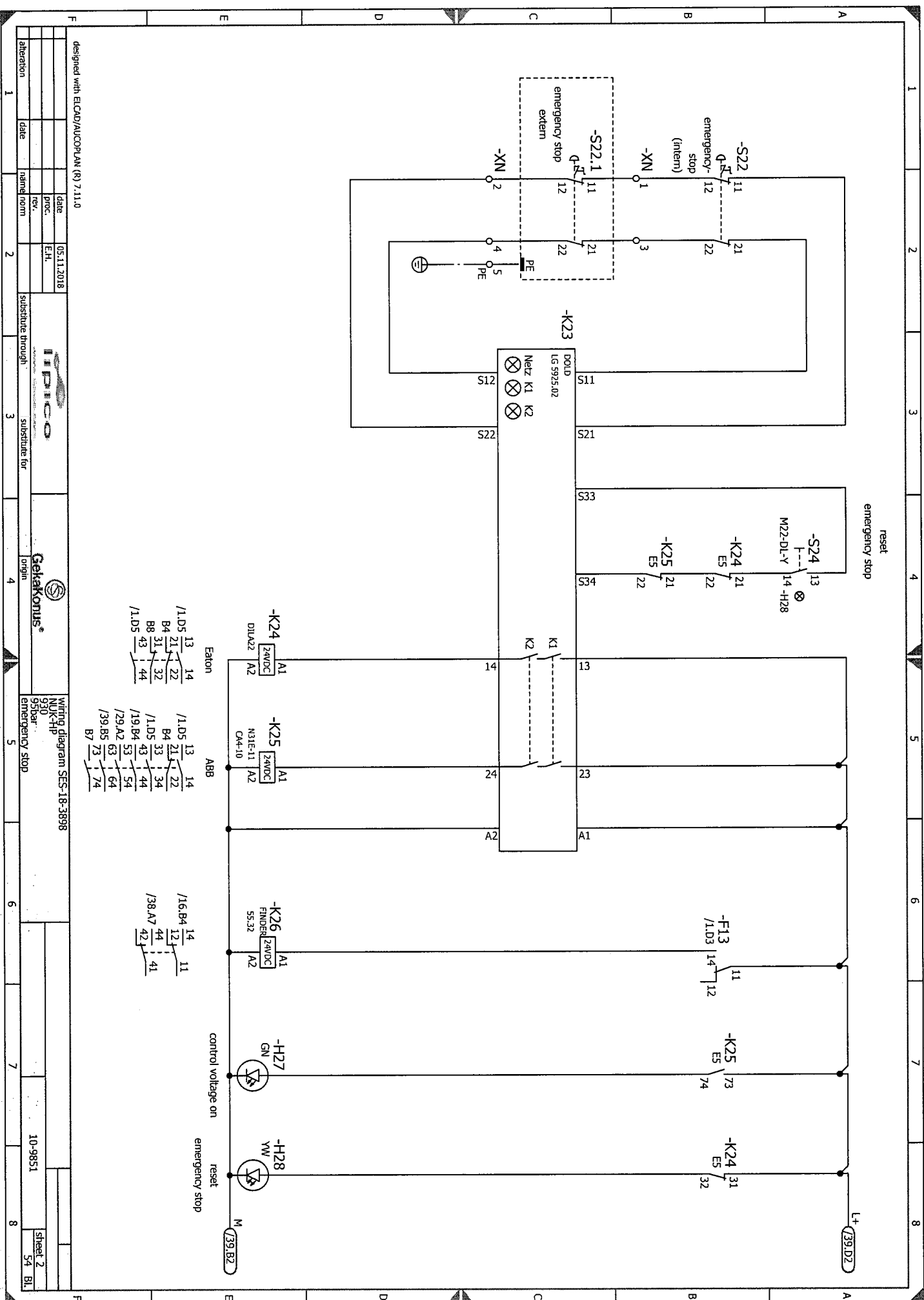
in operation

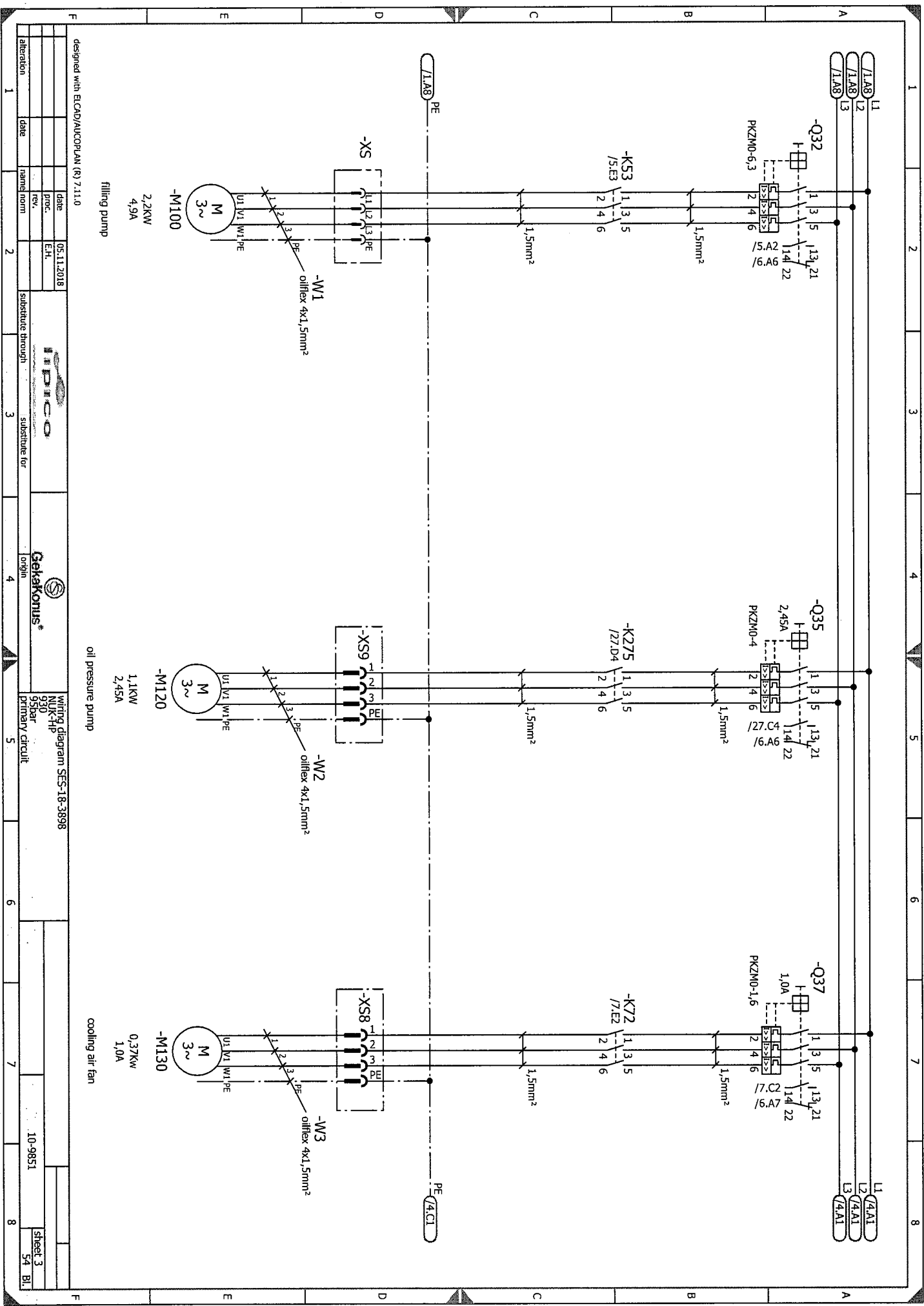
H57=extem fuel oil pump MFO

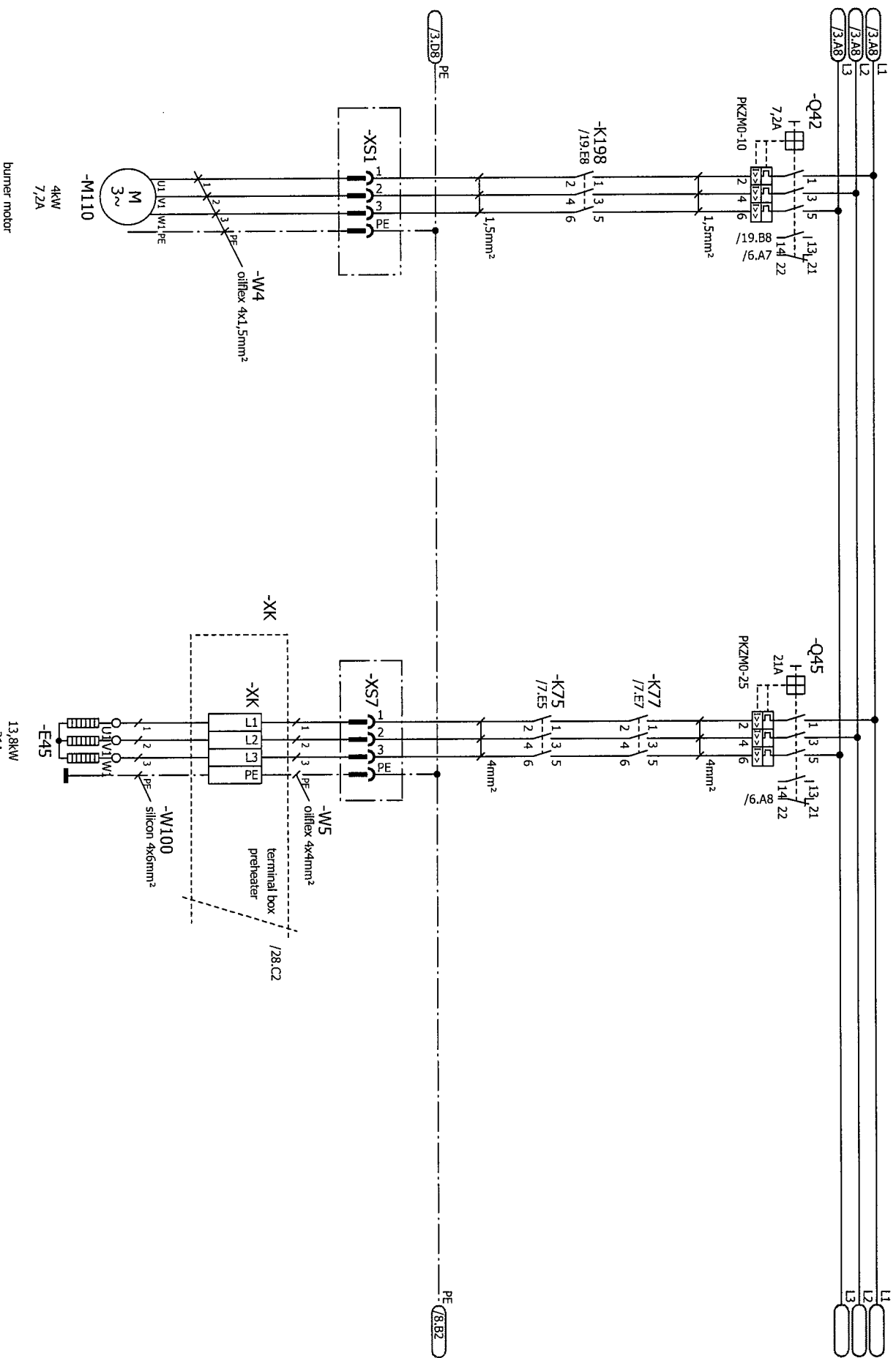
in operation

GekaKonus®

M182







designed with E/CAD/AUCOP/PLAN (R) 7.11.0

burner motor

4kW
7,2A

Wiring diagram SCS-18-3898

13,8kW
21A

preheater

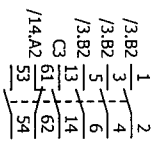
-W100
silicon 4x6mm²

terminal box
preheater

/28.C2

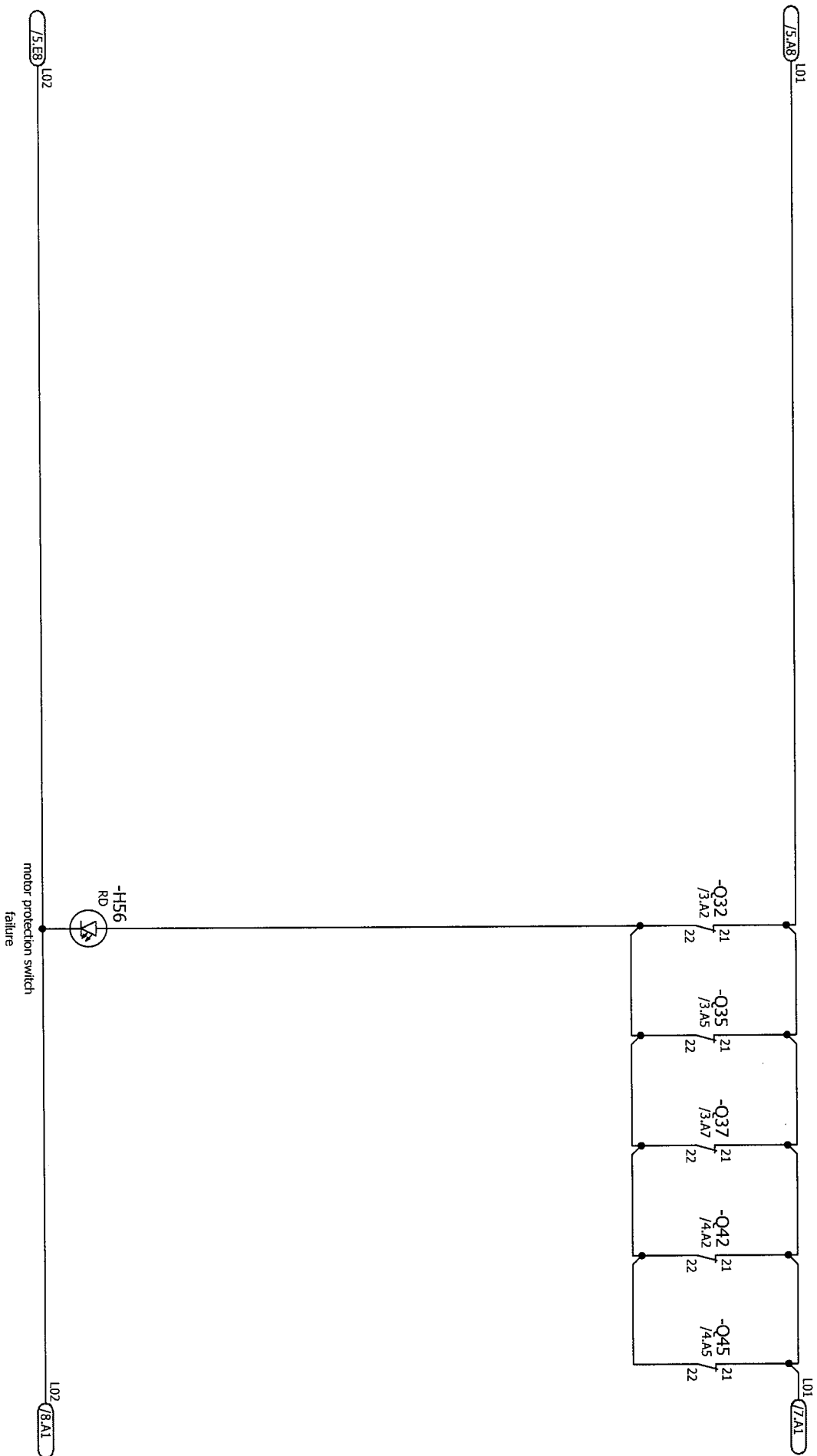
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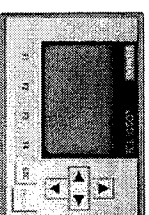
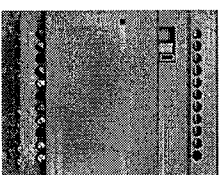
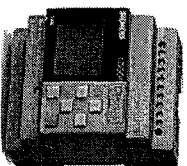


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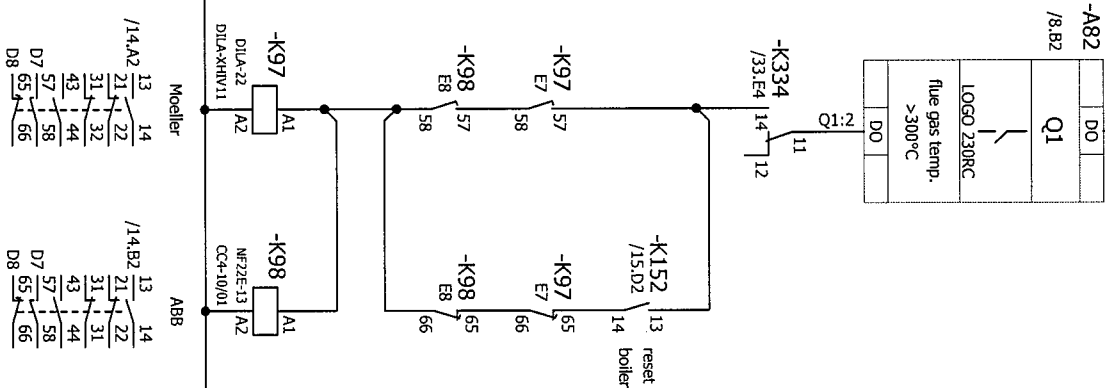
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designed with ELCAD/AUCOPLAN (R) 7.11.0		date		05.11.2018		proc.		E.H.		rev.		substitute through		substitute for		origin		writing diagram SES-18-3898		sheet 6		54 Bl.	
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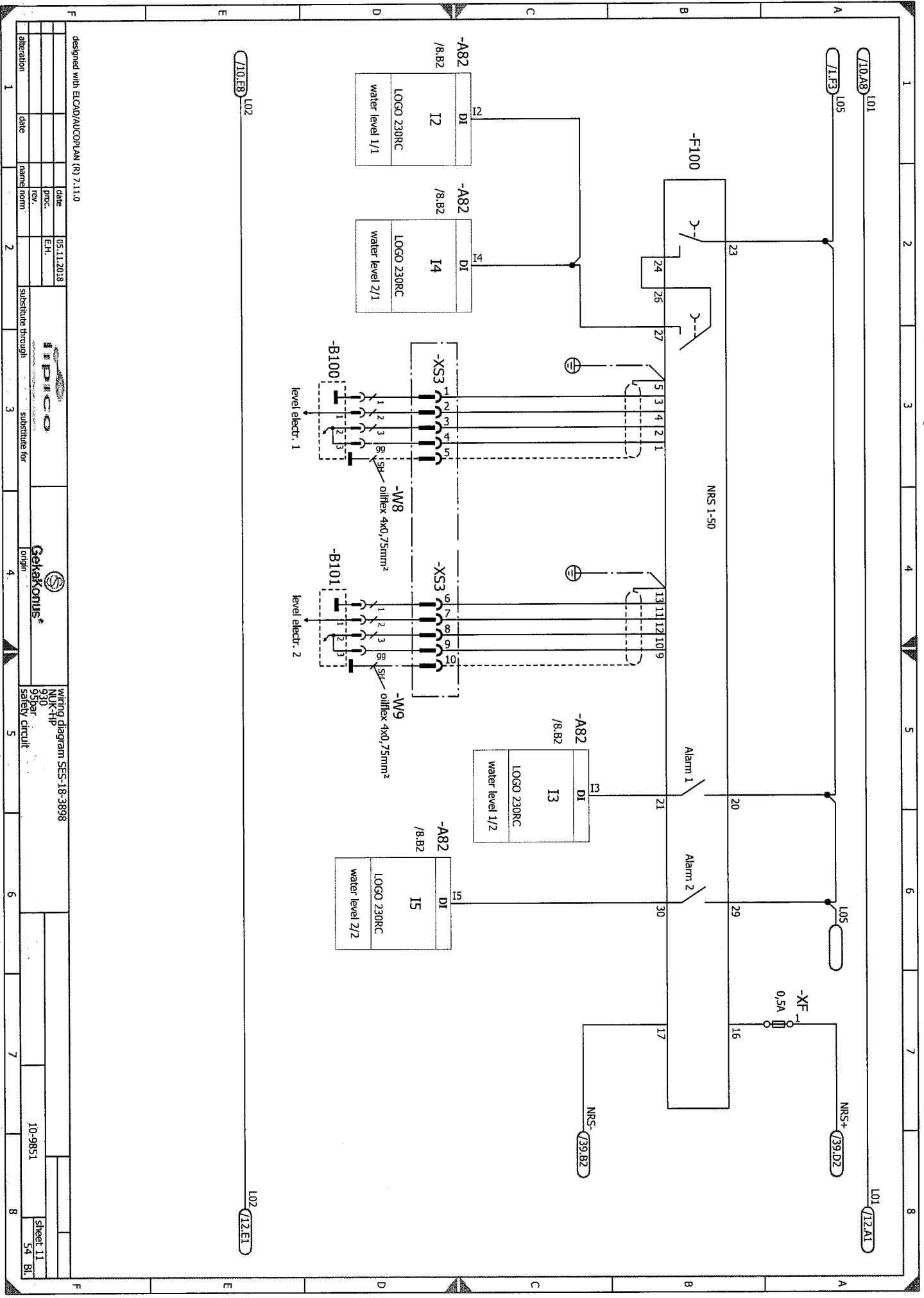


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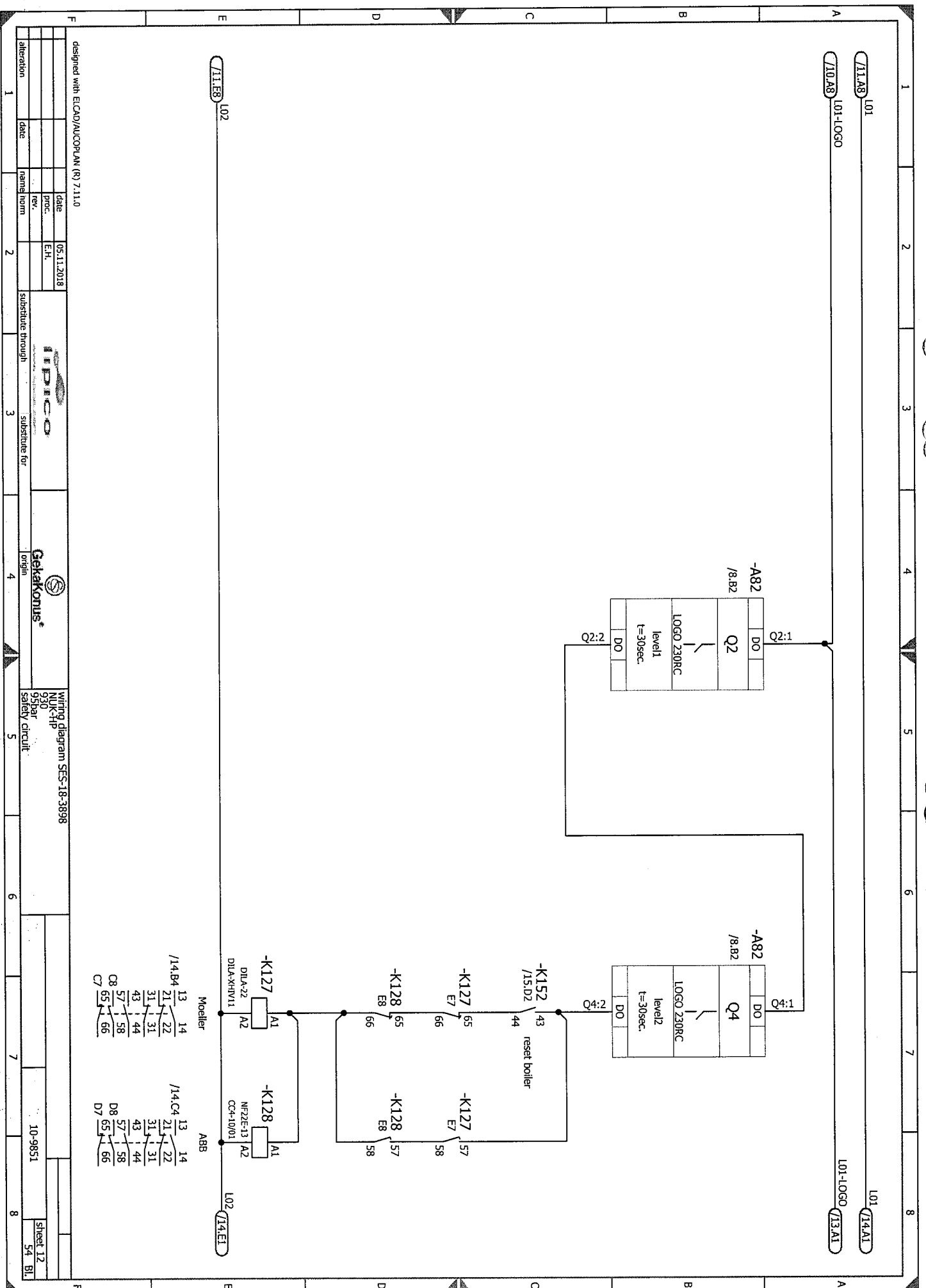
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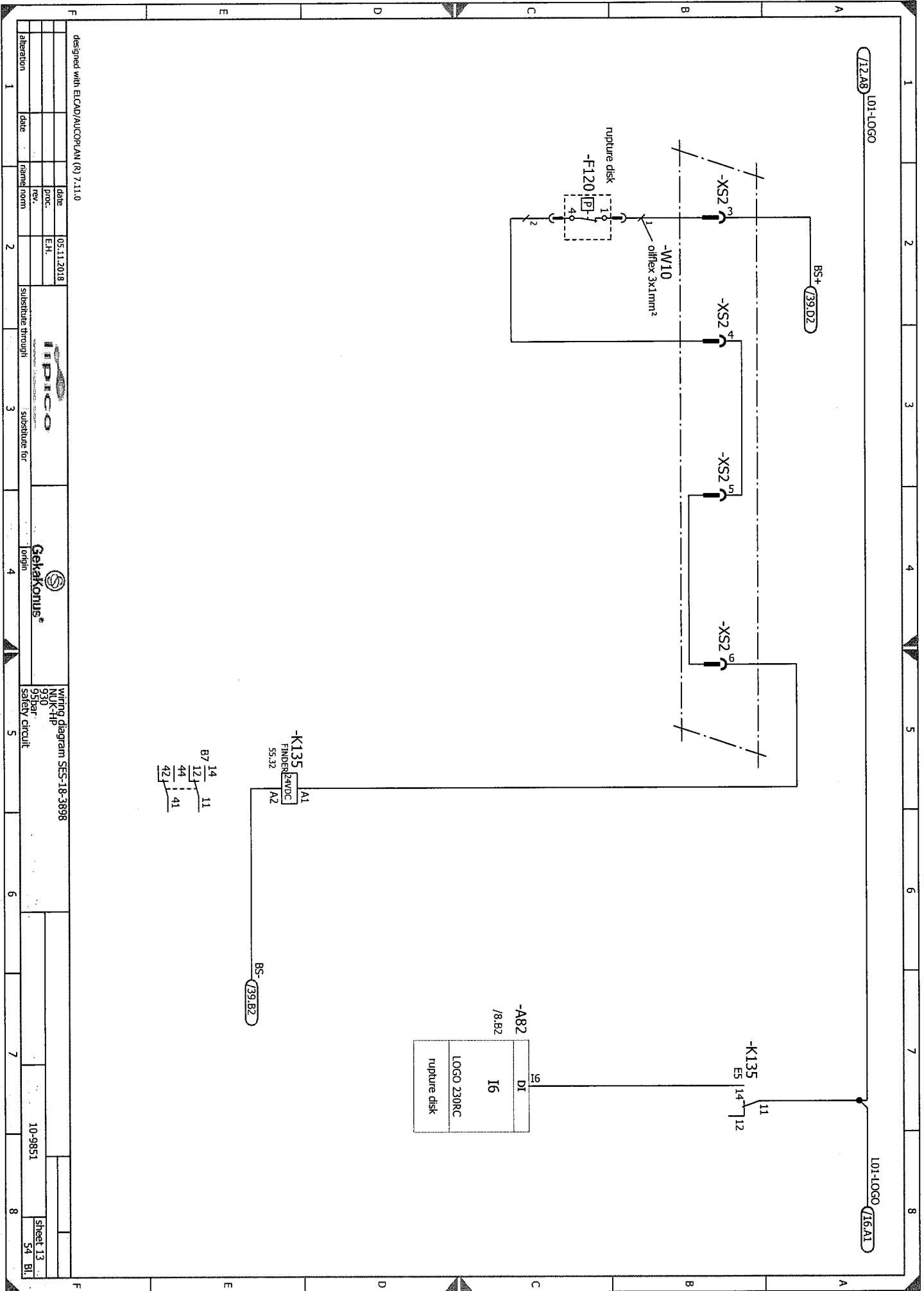
wiring diagram SES-18-3898
NUK-HP
930
95bar
cable circuit

sheet 9
54 RI

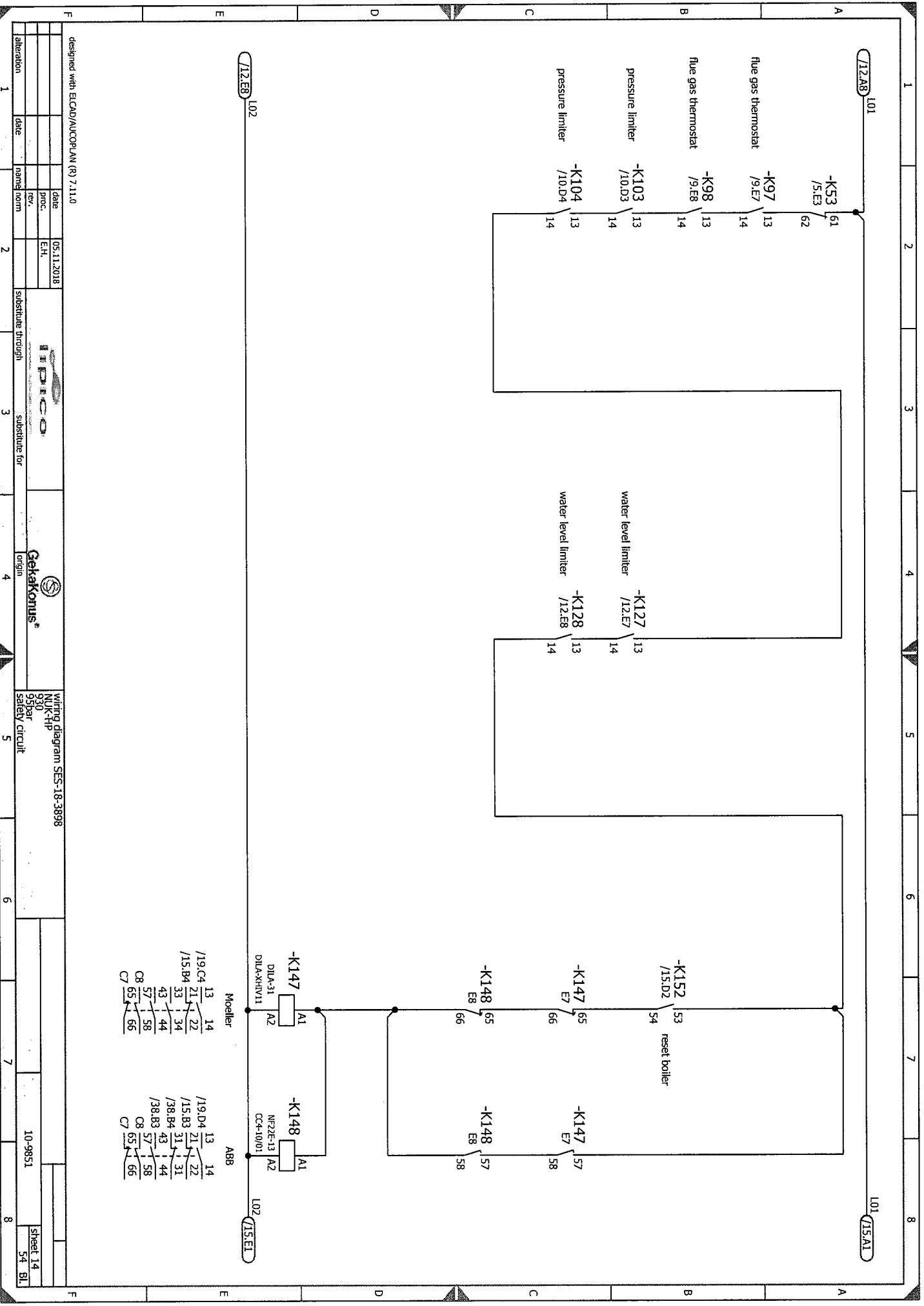


designed with EUCAD/AUCOPPLAN (R) 7.11.0		writing diagram SES-18-3898		sheet 11	
alteration	date	proc.	rev.	substitute through	substitute for
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alteration	date	name	room	substitute through	substitute for	origin	writing diagram SES-18-3898	10-9851	sheet 13
1	05.11.2018	E.H.					NK-41P		54 Bl.
		proc.					930		
		rev.					95W		



designed with ELCAD/AUCOPAN (R) 7.11.0

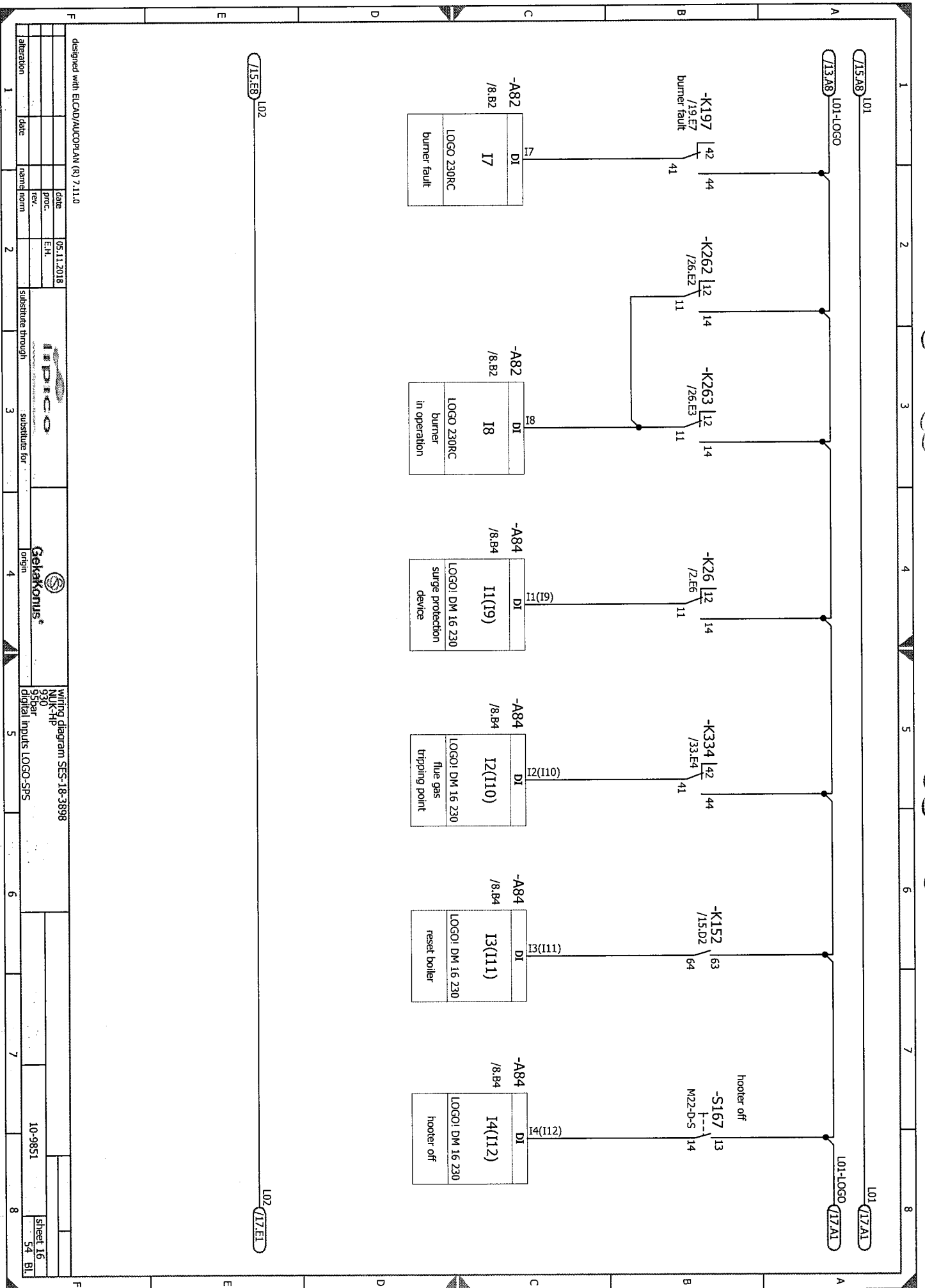


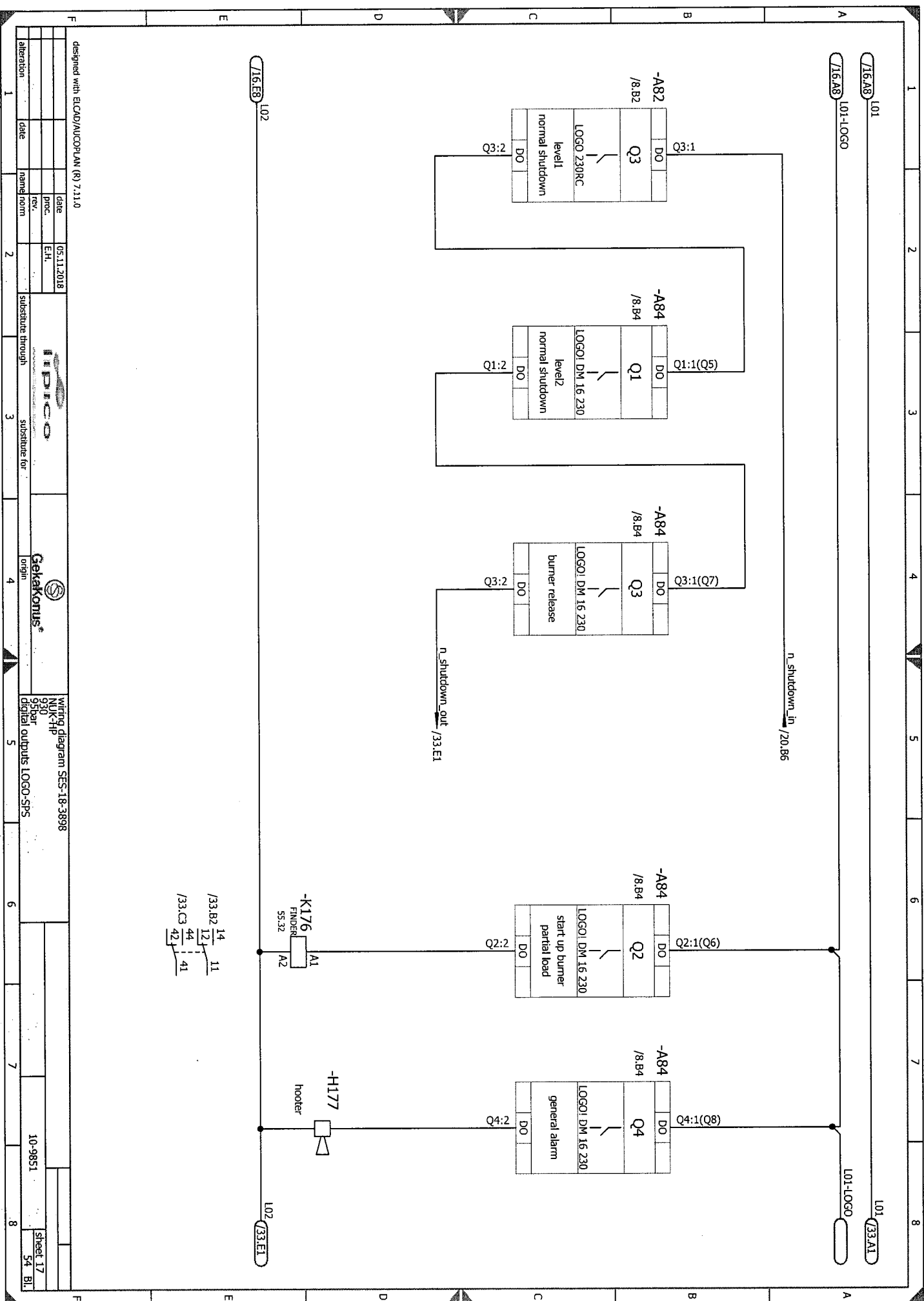
wiring diagram SES-18-3898

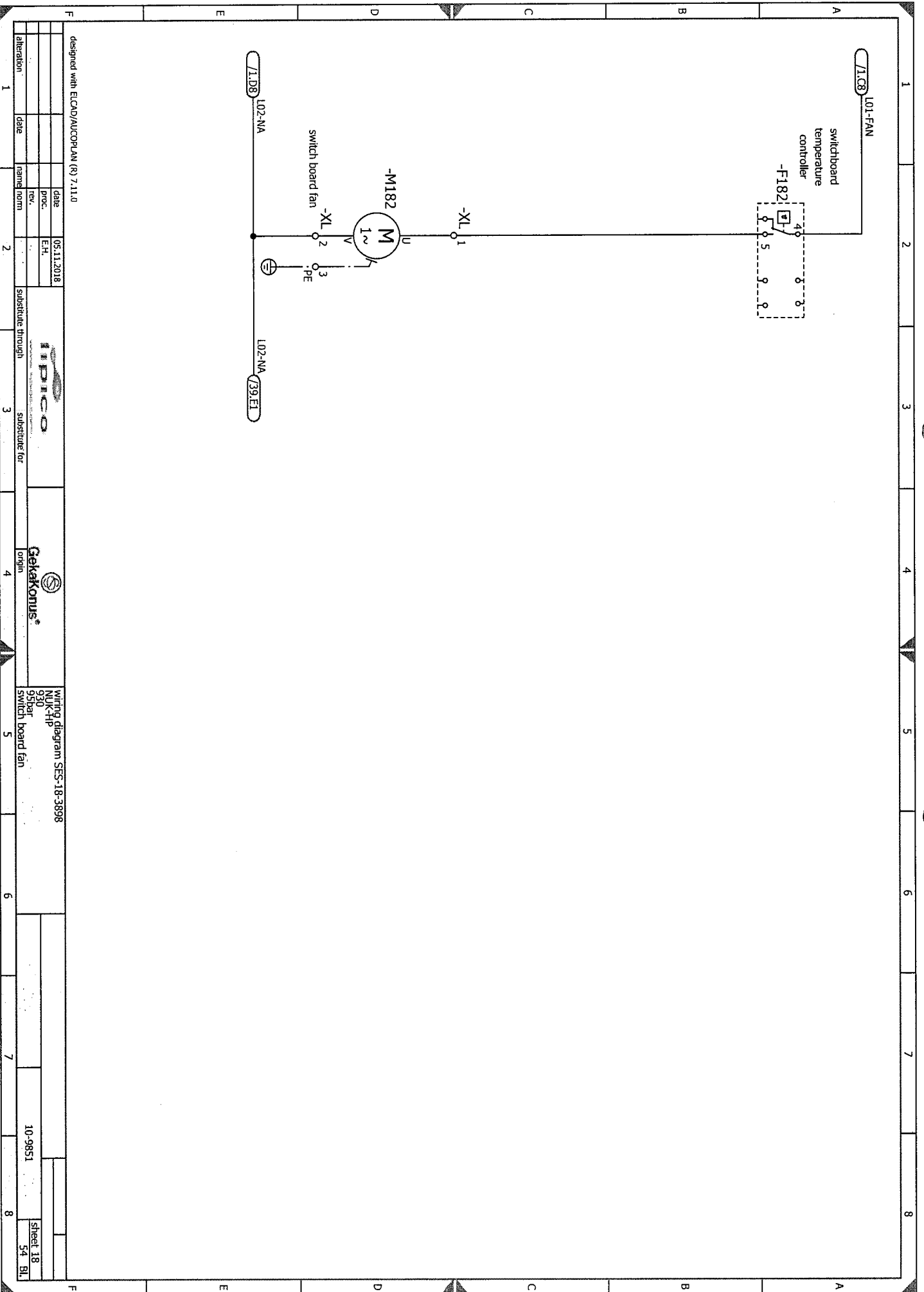
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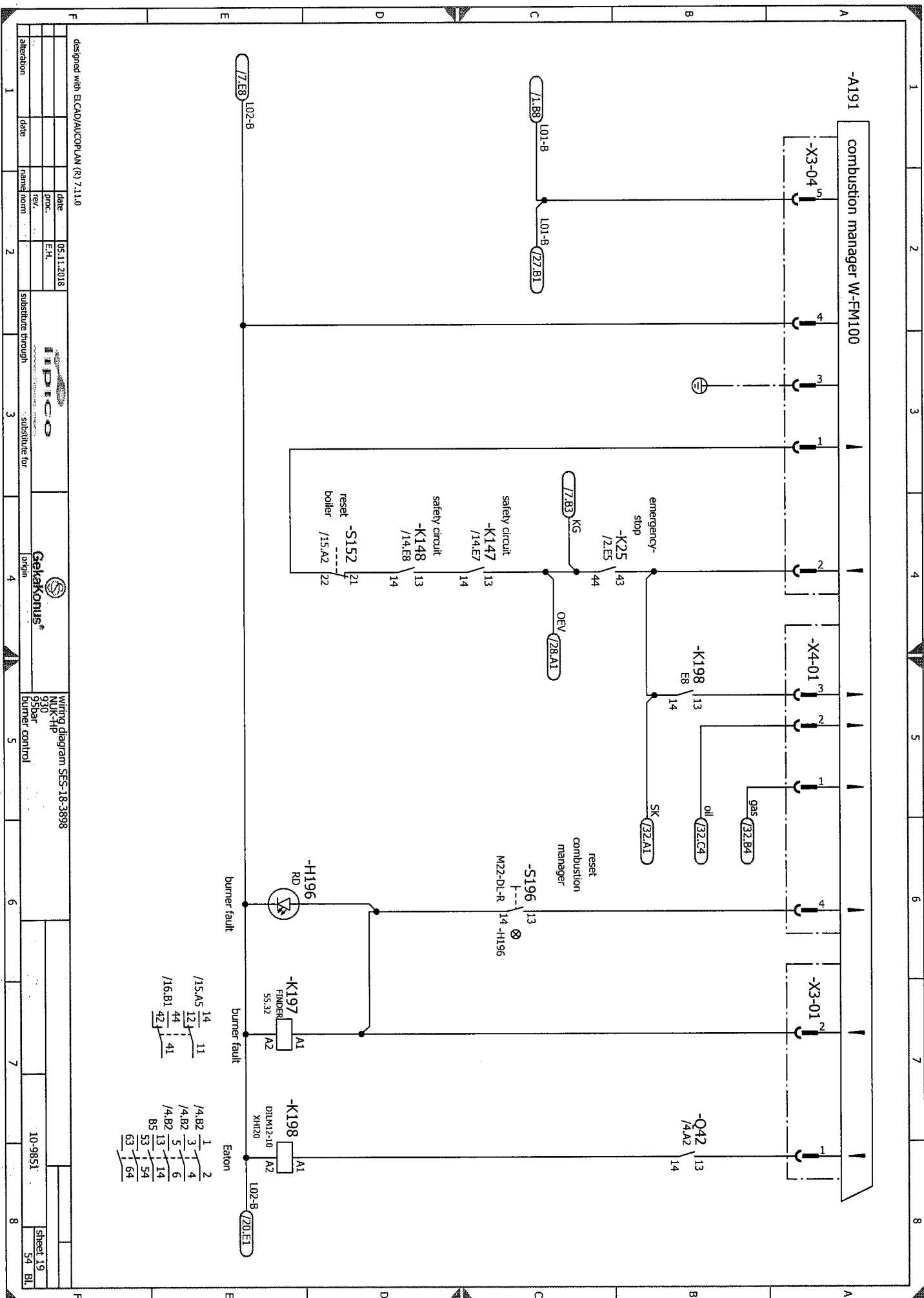
sheet 14

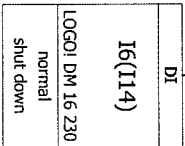
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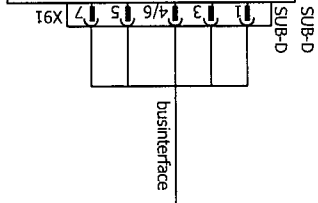


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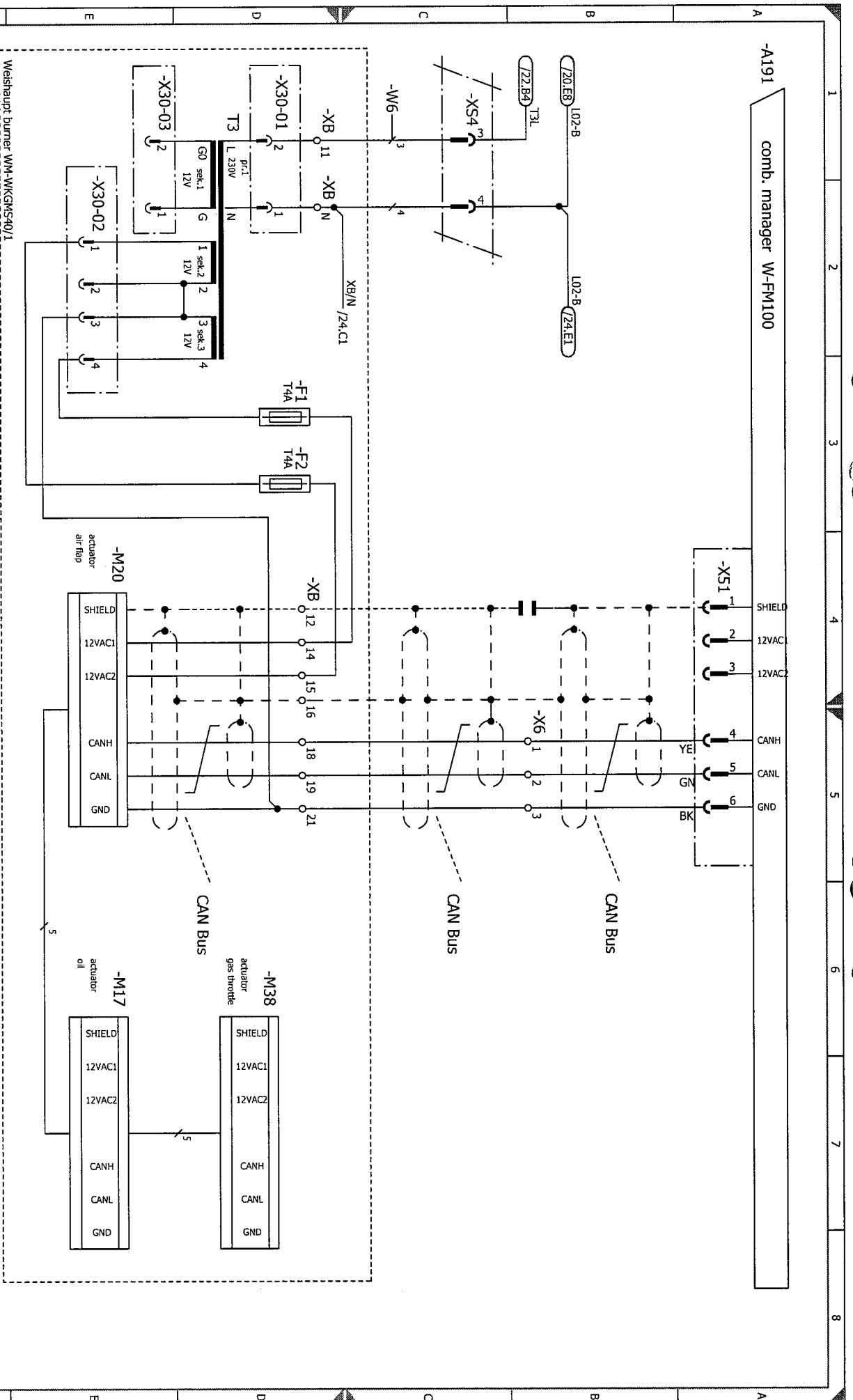
wiring diagram SES-18-3898

wiring diagram
NUK-HP
930
95bar
burner control

54 RI

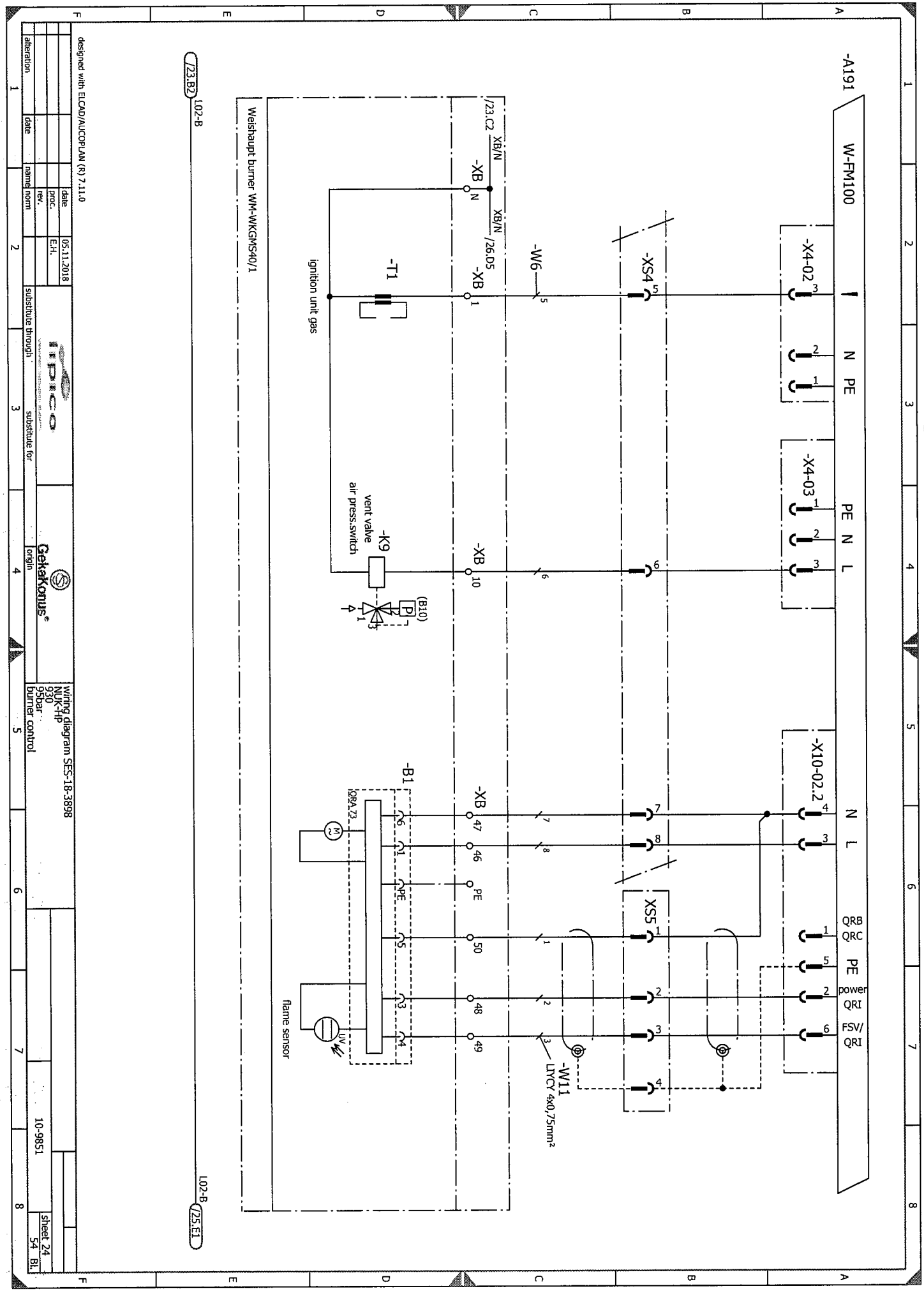


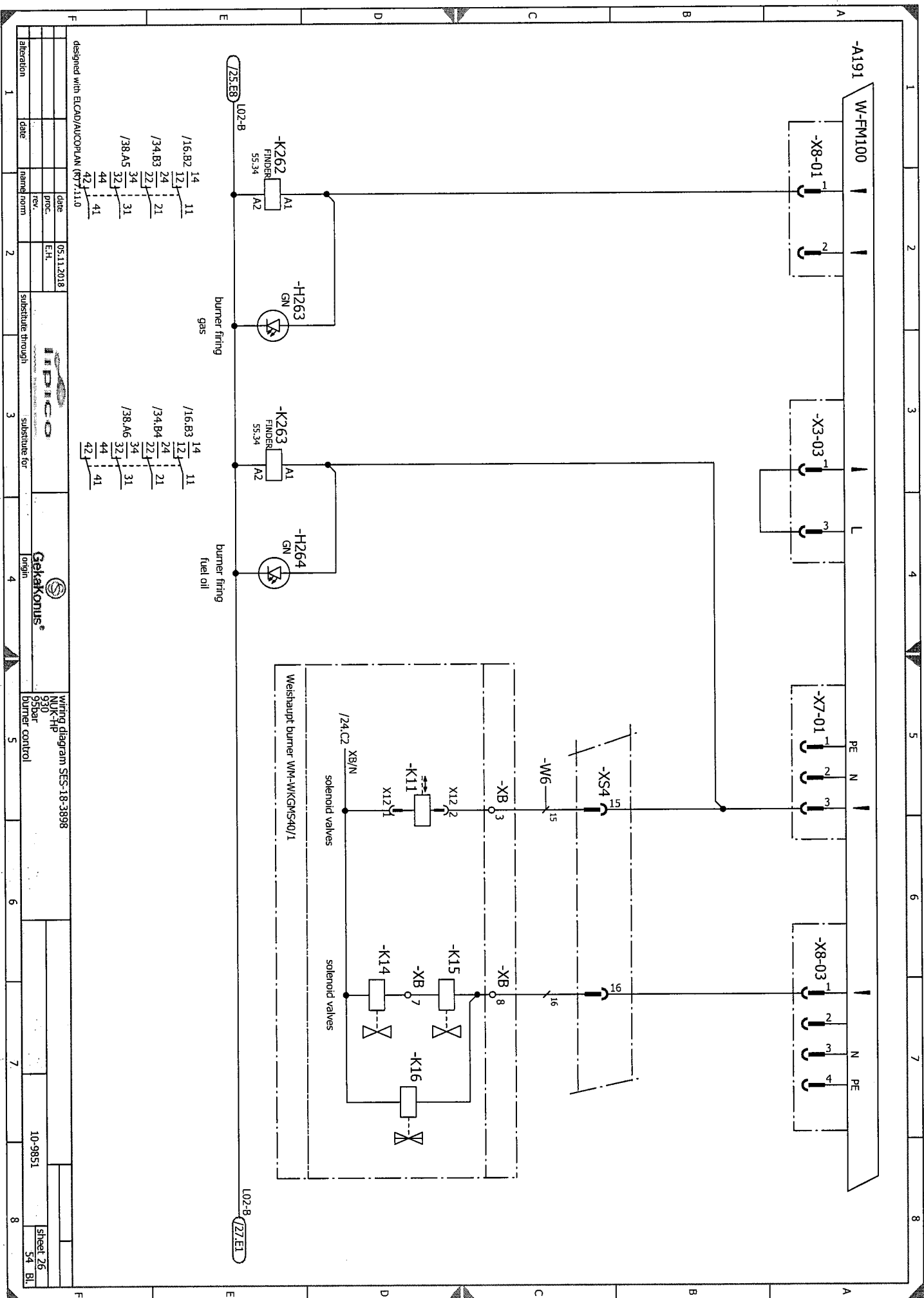
sheet 21	54 Bl.
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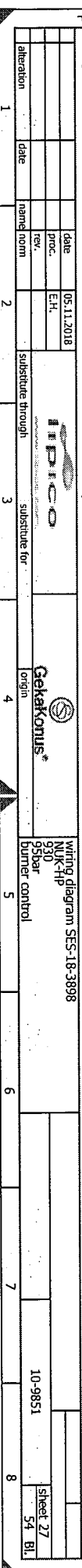


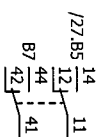
designed with ELCAD/AUCOPLAN (R) 7.11.0

alteration		date	proc.	rev.	name	nom	substitute through	substitute for	origin	writing diagram SES-18-3898		930	950	10-9851	sheet 23
1										Gekakomus					54 Bl.
2		05.11.2018	E.H.							writing diagram SES-18-3898					
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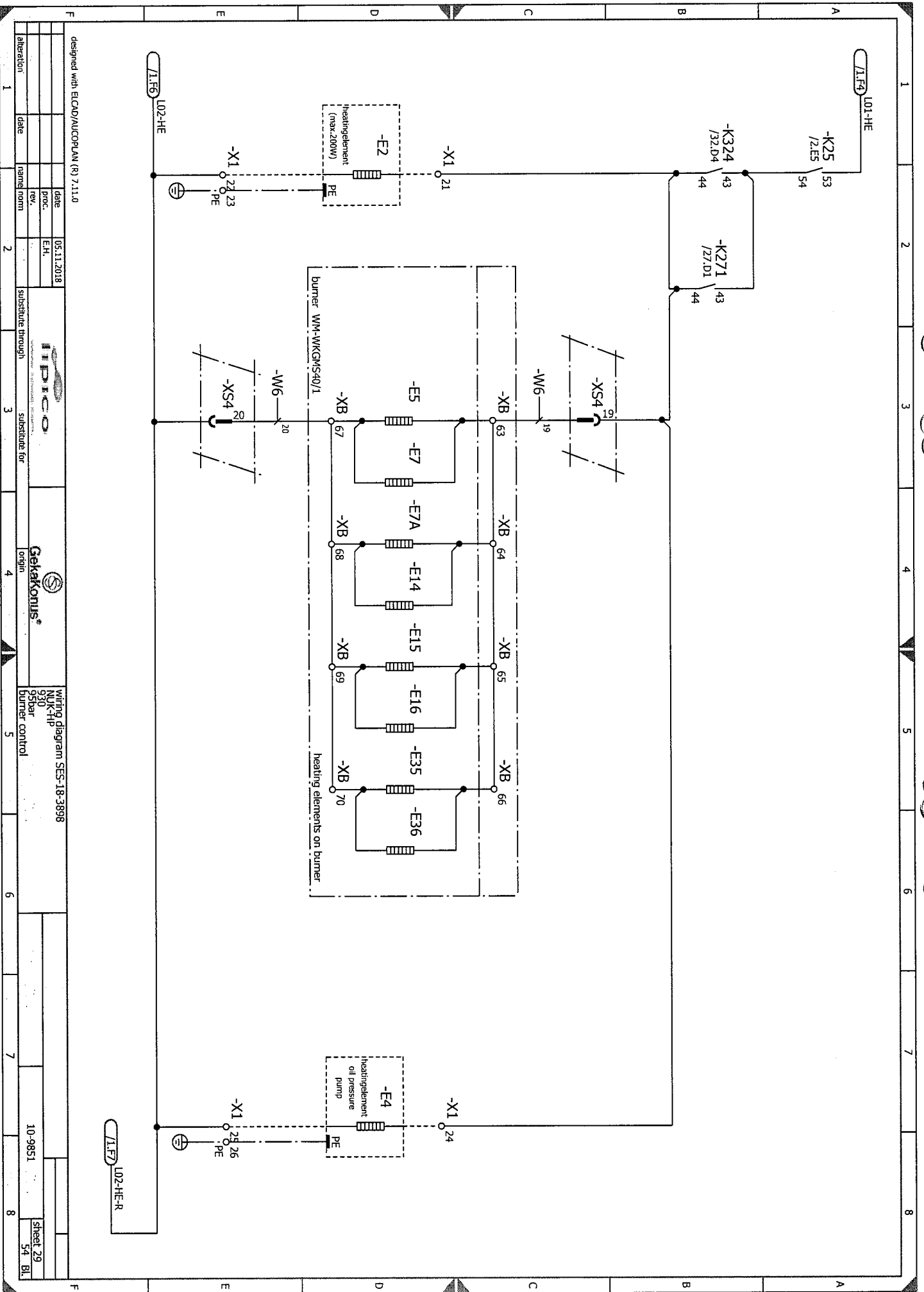


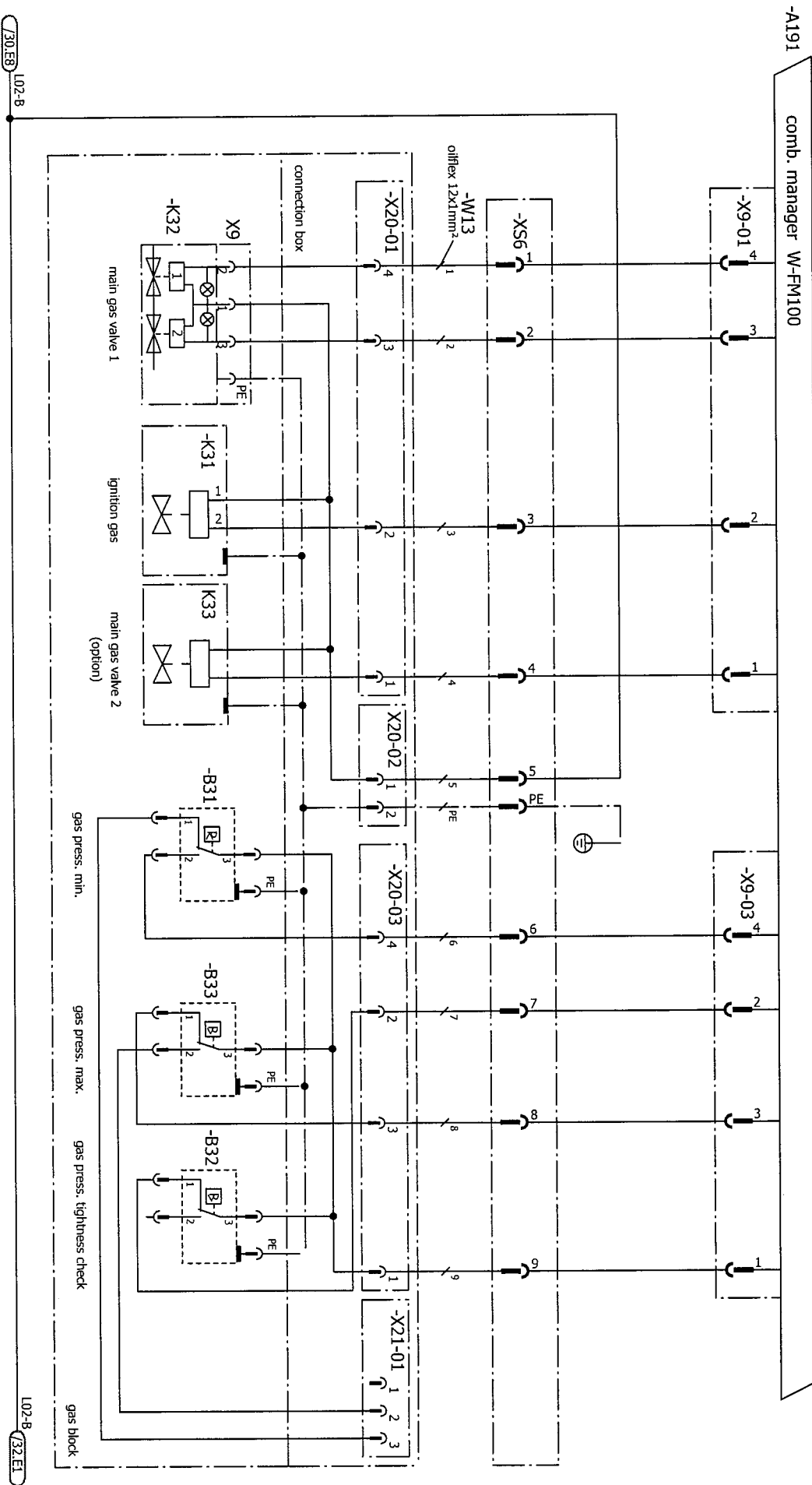




wiring diagram SES-18-3898
NUK-HP
930
95bar
burner control

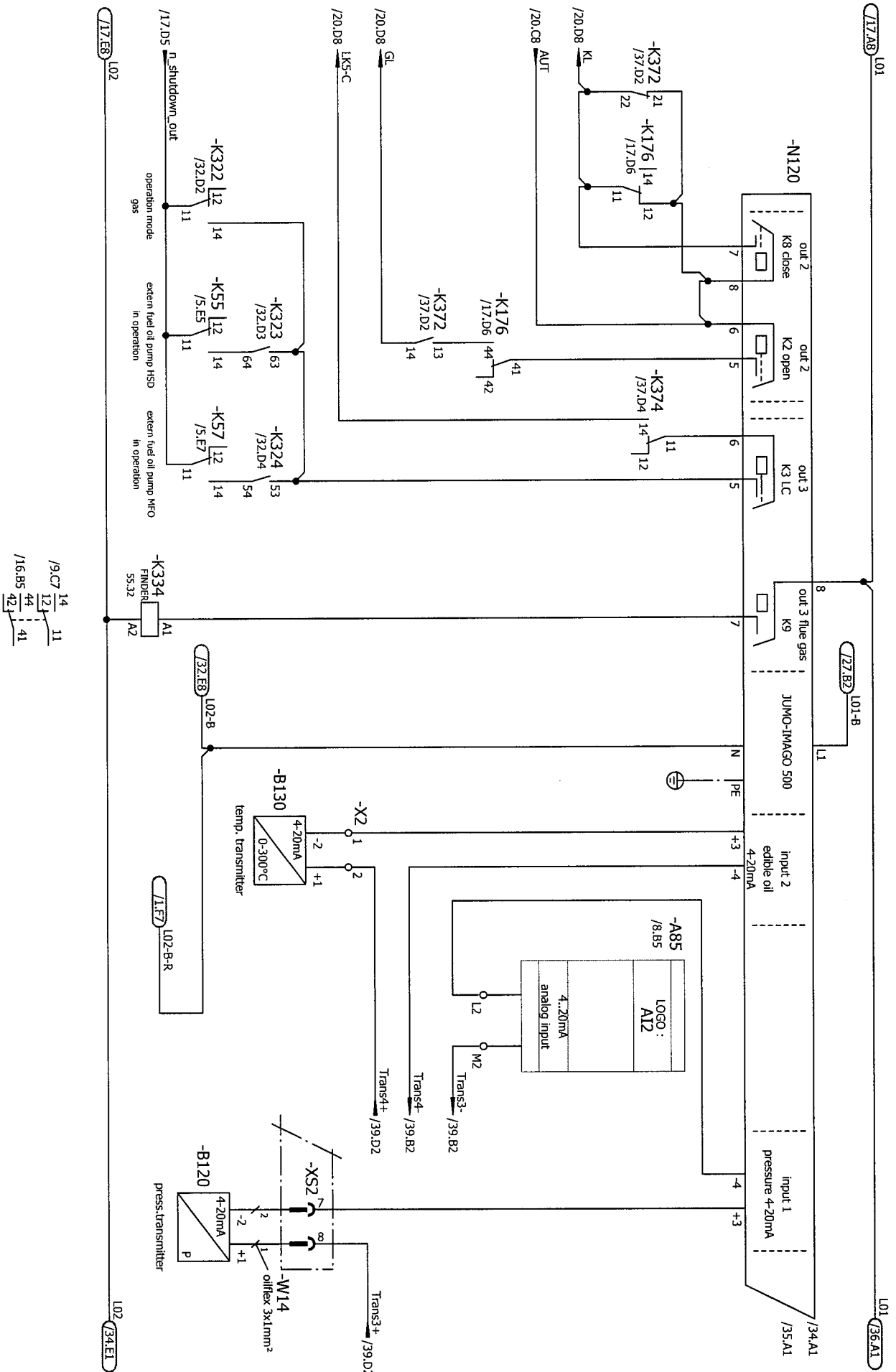
sheet 28
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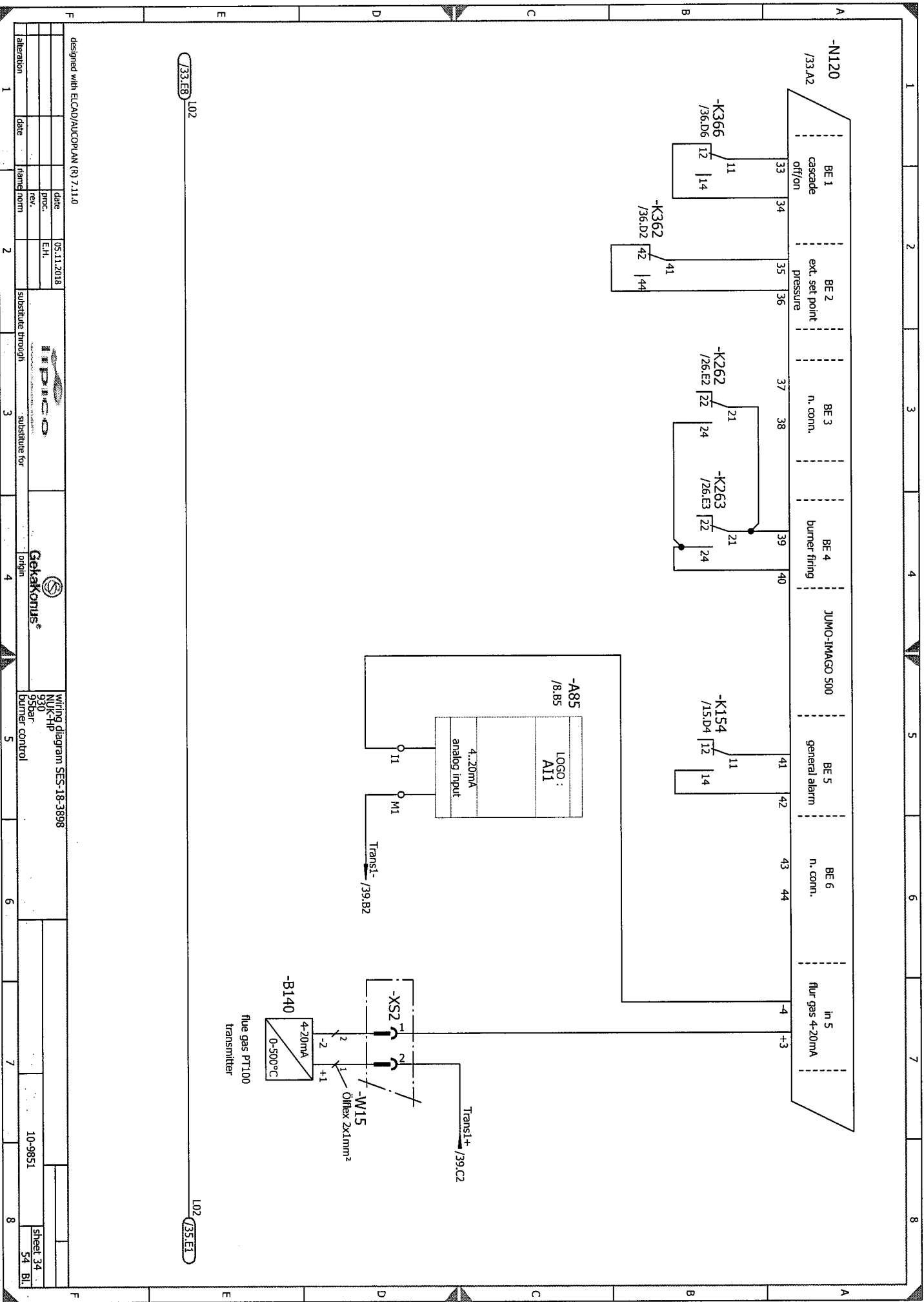
designed with ELCAD/AUCOPLAN (R) 7.11.0

date	05.11.2018	substitute through	substitute for	origin	wiring diagram SES-18-3898	sheet 31
proc.	E.H.				NUK-HP	54 Bl.
rev.					95bar	
alteration	date	name nom			burner control	
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designed with ELCAD/AUCOPLAN (R) 7.11.10

alteration		date	05.11.2018	substitute through		substitute for	origin	writing diagram SES-18-3898		NUK-HP 95bar burner control		10-9851		sheet 33		54 Bl.	
1				2				3		4		5		6		7	
8		9		10		11		12		13		14		15		16	
17		18		19		20		21		22		23		24		25	
26		27		28		29		30		31		32		33		34	
35		36		37		38		39		40		41		42		43	
44		45		46		47		48		49		50		51		52	
53		54		55		56		57		58		59		60		61	
62		63		64		65		66		67		68		69		70	
71		72		73		74		75		76		77		78		79	
80		81		82		83		84		85		86		87		88	
89		90		91		92		93		94		95		96		97	
98		99		100		101		102		103		104		105		106	



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date 05.11.2018

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alteration

substitute through

origin

burner control

wiring diagram SES-18-3898

NLK-HP

930

95bar

10-9851

sheet 34

54 Bl.

33.E8

33.E1

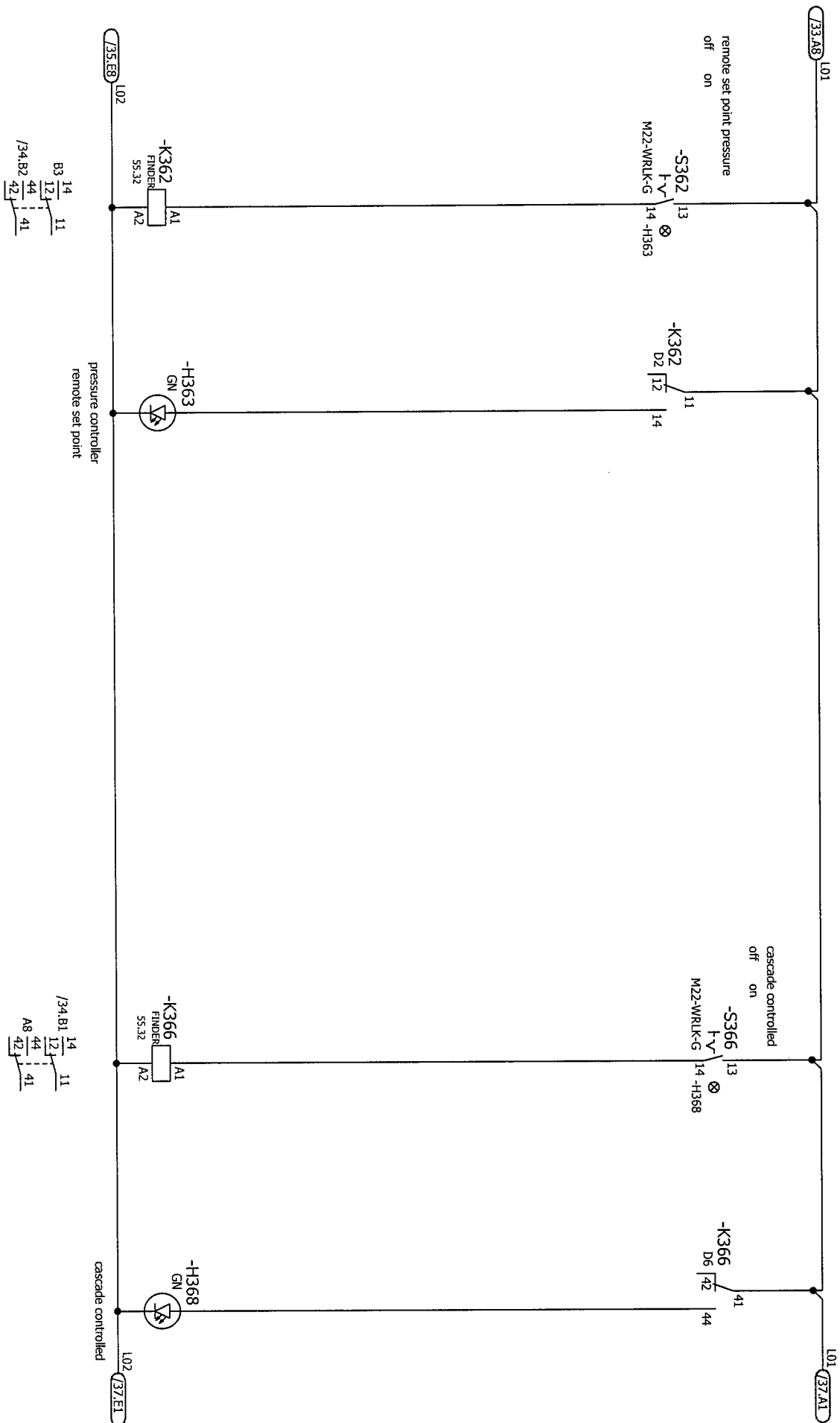
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102



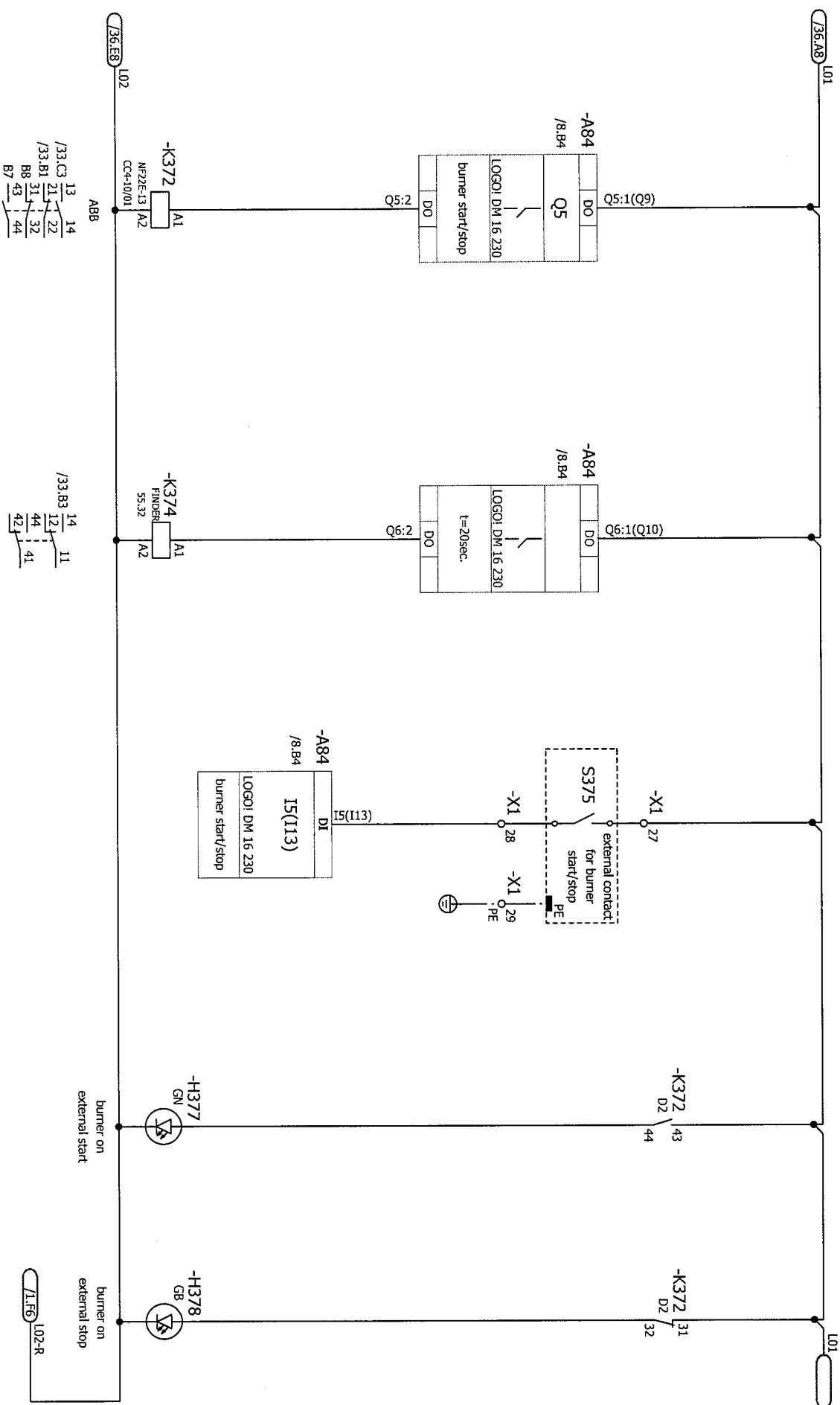
wiring diagram SES-18-3898
NUK-HP
930
95bar
burner control

sheet 35
54 Bl.



designed with ELCAD/AUCOPLAN (R) 7.11.0

date		05.11.2018	date		05.11.2018
proc.		E.H.	proc.		E.H.
rev.			rev.		
alteration			alteration		
date			date		
name		room	name		room
substitute through			substitute through		
substitute for			substitute for		
origin			origin		
writing diagram SES-18-3898			writing diagram SES-18-3898		
NUK-HP			NUK-HP		
95bar			95bar		
burner control			burner control		
10-9851			10-9851		
sheet 36			sheet 36		
54 Bl.			54 Bl.		

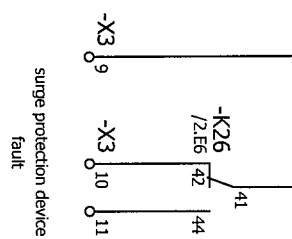
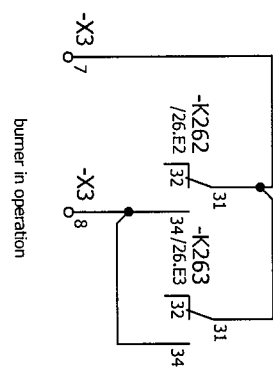
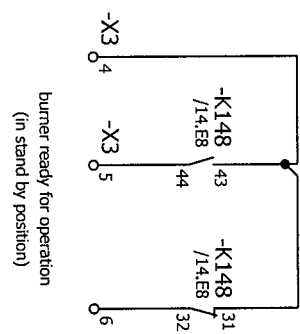
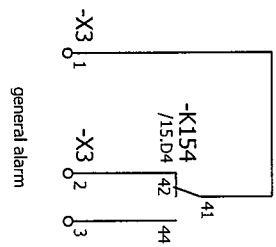


13 14
/33.C3 21 22
/33.B1 31 32
B8 43 44
B7

14 11
/33.B3 12 11
44 41
42

designed with ELCAD/AUCOPLAN (R) 7.11.0

alteration		date	05.11.2018	substitute through		substitute for		origin		writing diagram SES-18-3898		10-9851		sheet 37	
1		date		2		3		4		5		6		7	
		proc.	E.H.												
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		date	05.11.2018			
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		rev.				
alteration	date	naming	nom	substitute through	substitute for	
				 INSTRUMENTATION & PROCESS CONTROL		
				 Gekkonus® GELATINE FILMS FOR X-RAY DETECTION		
				writing diagram SES-18-3898 NIK-TIP 930 95bar		
				pdt. free cont.		
				10-9851		
				sheet 38 54 Bl.		

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alteration	date	name	norm	date	proc.	rev.	E.H.	substitute through	substitute for	origin	writing diagram SES-18-3898	930 NUK-HP	-XS4, -XS5, -XS6	10-9851	sheet 44	54 Bl.											
																	cable mark extern		target mark extern	connection strip				target mark intern	cable mark intern		appliance comment
																			appliance mark	terminal number	type	wire connection	terminal comment	representation place	appliance mark	connection mark	
appliance comment																											
strip terminal : -XS4																											
totality : 22																											
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